

**DEPARTMENT OF COMMUNITY HEALTH
FACULTY OF MEDICINE
ADDIS ABABA UNIVERSITY**

COMH 603

Principles of Epidemiology
(3 credits)

LECTURE NOTE

by

YEMANE BERHANE

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Community Health Department
Faculty of Medicine
Addis Ababa University

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I. INTRODUCTION TO EPIDEMIOLOGY

Epidemiology is considered as the basic science of public health. It provides useful tools and methods to describe variations in disease occurrence and identify factors that influence the occurrence of disease among population groups. The occurrence of disease is dependent on variations in exposure of individuals in the population to the causes of the disease that are commonly behavioural and environmental. These facts have been known and some important environmental exposures that influence disease occurrence identified since the time of Hippocrates. The importance of epidemiology has become increasingly important in modern public health practice, thus the data collection and analytical techniques are being constantly revised to meet the challenges of obtaining the necessary information for proper planning of health interventions. Basic laboratory researches are important in advancing our biologic understanding of diseases but the quantification of the magnitude of the exposure-disease relationship in human need to be determined using epidemiology in order to alter the risk through effective interventions; in fact epidemiology has provided useful information that has formed the basis for some revolutionary public health decisions long before the basic mechanism of those diseases was understood.

Epidemiology studies the nature of diseases, and their causes, and it uses systematic methods of measurement to test ideas, questions and hypotheses, and hence it is a science (bio-science), serving medicine and public health. Epidemiologists are not always occupied in theoretical application of the discipline; some are occupied in applying available knowledge in public health practice to achieve better health conditions. This applied work though requires scientific evidences for proper planning and evaluation it is not science. Thus, there are scientists and practitioners in the field of epidemiology.

A. Definition

Epidemiology is the study of the frequency, distribution, and determinants of health-related states or events in specified populations, and the application of this study to the control of health problems.

The definition emphasizes that epidemiology is concerned with the collective health of individuals in communities and in getting appropriate solution to alleviate the health problems. It is used to describe the health problem (what is occurring) and its frequency (how many); who is affected, where, when; why diseases are occurring; and how we can influence the occurrence (what type of intervention). The unique contribution of epidemiology to health sciences is the fact that epidemiological studies are conducted in human populations. Thus, due to ethical and practical reasons epidemiological studies are largely observational.

B. Basic Epidemiologic Assumptions

In order to fully grasp the notions of epidemiology it is important to understand the two basic assumptions in epidemiology:

1. Human disease does not occur at random: there are patterns of occurrence in which some behavioural and environmental factors (exposures) increase the risk of acquiring/developing a particular disease among group of individuals.
2. Human disease has causal and preventive factors that can be identified through systematic investigation of populations or group of individuals within a population in different places or at different times. Thus, identifying these factors creates opportunity for prevention and control of diseases in human population either by eliminating the cause or introducing appropriate treatment.

C. Scope/use of epidemiology

The use of epidemiology in advancing health sciences and improving public health practices has been greatly expanded in the last few decades. Its scope at the beginning was limited to understanding epidemics. Now it is the basis of advancing our understanding of all kinds of diseases whether they belong to communicable, non-communicable or injury category. It is used in laboratory sciences, clinical medicine and public health. Its scope in public health ranges from *routine surveillance to research strategies* for the testing of hypotheses about causes, measurement of health and disease risks and evaluations of preventive, diagnostic and therapeutic programmes and technologies. Epidemiology is also a collection of *applied disciplines*, i.e., every disease entity has its own epidemiology (infectious, cardio-vascular, cancer, etc.). Other studies focus on health risks (occupation, smoking, diet, social conditions, etc.). Some of the uses of epidemiology in public health practice are mentioned below:

1. Elucidate the natural history of disease.
2. Describe the health status of the population.
3. Establish causation of disease.
4. Provide understanding of what causes or sustains disease in populations.
5. Define standards and ranges for normal values of biological and social measures.
6. Guide health and healthcare policy and planning.
7. Assist in the management and care of health and disease in individuals.

8. Evaluate the effectiveness of intervention.

D. Major categories of epidemiology

Epidemiology can be categorized into two:

1. Descriptive Epidemiology - Defines the amount and distribution of health problems in relation to person, place and time. It answers the questions ***who, where and when.***
2. Analytic Epidemiology – involves explicit comparison of groups of individuals to identify determinants of health and diseases. It answers the questions ***why and how.***

A further detail on each category is given in the chapter dealing with types of epidemiological studies. It is important to note that each has its own unique strengths and limitations. The choice of a particular study type is dependent on the purpose of the study. Thus, it is critical to gain a thorough knowledge of each of them in order to make rational decision in choosing appropriate methods.

E. History of Epidemiology

Although epidemiological thinking has been traced to the time of Hippocrates, the discipline did not flourish as an independent discipline until the 20th century. Some key dates and contributions to the development of epidemiologic thinking and methods include:

460 B.C – Hippocrates, the father of modern medicine. For the first time in the fifth century B.C. he suggested that the development of human disease might be related to the external as well as personal environmental of an individual.

1662 - John Graunt published *Natural and Political Observations on the Bills of Mortality*. He was the first to quantify patterns of birth, death and disease occurrence, noting male-female disparities, high infant mortality, urban-rural differences, and seasonal variations.

1747 - Lind used an "experimental" approach to prove the cause of scurvy by showing it could be treated effectively with fresh fruit.

1787-1872. Pierre Charles Alexandre Louis, sometimes called the "Father of Epidemiology", systematized the application of numerical thinking ("*la methode numerique*") and championed its cause. Using quantitative reasoning, he demonstrated that bloodletting was not efficacious therapy, and wrote books on tuberculosis and typhoid. Louis' influence was widespread, primarily through his students. William Farr, William Guy, William Budd (all students of Louis) – founded the Statistical Society of London.

1839 - William Farr took responsibility for medical statistics in the Office of the Registrar General for England and Wales. He extended the epidemiologic analysis of morbidity and mortality data, looking at effects of marital status, occupation, and altitude.

1854 - John Snow demonstrated that the risk of mortality due to cholera was related to the drinking water provided by a particular supplier in London. He used a "natural experiment" to test his hypothesis. In another study conducted by Snow in 1854, he linked an epidemic of cholera to a specific pump, the "Broad Street Pump". According to literatures, Snow removed the handle of that pump and aborted the cholera epidemic.

1937 Austin Bradford Hill, mainly worked on the Principles of Medical Statistics and suggested the criteria for establishing causation.

1950's-1970's. Major epidemiology successes in the area of – fluoride, tobacco, blood pressure and stroke, CHD risk factors, toxic shock syndrome, Legionnaire's disease, Reye's syndrome, endometrial cancer and exogenous estrogens.

Originally epidemiology was concerned with epidemics of communicable disease. Lately, epidemiology was extended to endemic communicable diseases. More recently, epidemiologic methods have been applied to chronic diseases, injuries, birth defects, maternal and child health, occupational health, and environmental health. Now, even health behaviours, such as care-seeking, safety practices, violence, and hygienic practices are valid subjects for epidemiologic investigation.

Some of the important factors that lead to progressive development of epidemiology are:

- The need for quantitative reasoning in public health
- Possibility of conducting comparative studies – comparison of groups or populations
- Increasing availability of vital statistics system
- Hygienic and public health movement
- Improvements in diagnosis and classification
- Advances in the field of statistics
- Advances in computer applications and development of user-friendly statistical software
- Increasing availability of personal computers
- Biotechnology revolution
- Advances in genomics

II. COMMUNICABLE (INFECTIOUS) DISEASE EPIDEMIOLOGY

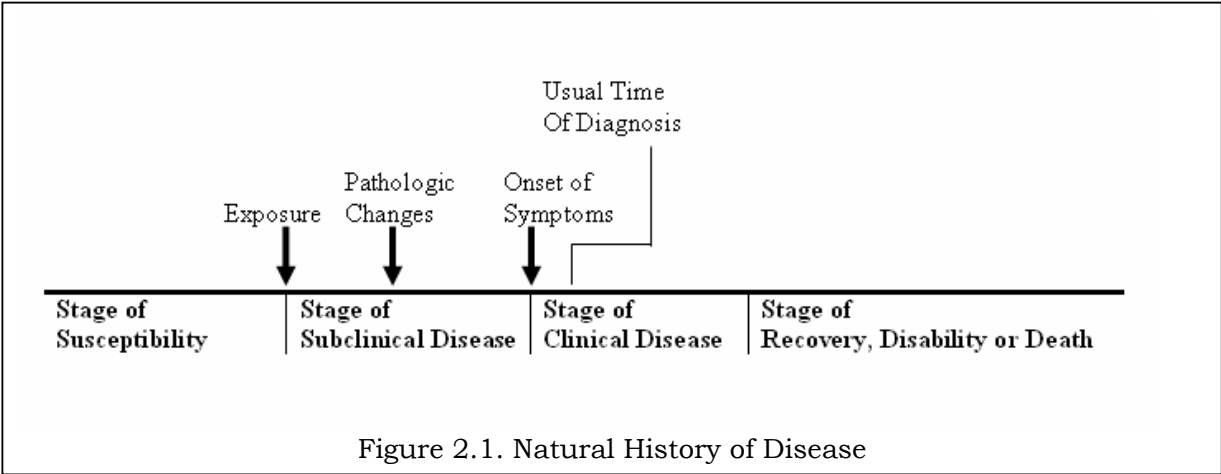
A. Introduction

Despite the great scientific advances that have reduced morbidity and mortality from communicable diseases over the past decades, communicable diseases continue to account for a major proportion of acute illnesses, even in technologically advanced countries, though the types of diseases may vary from place to place. Some important aspects of infectious diseases are discussed below. This group of diseases are characterized by the presence of the infectious agent in addition to susceptible human population. Transmission from one host to another is fundamental to the survival of infectious agent, since any host will eventually either clear the infection or die, even that is from unrelated cause. Although most methods used in general epidemiology are applicable to the study of infectious diseases, additional concepts that are described in this section are needed.

B. Natural History of Diseases

The natural history of disease refers to the progression of a disease process in an individual over time, in the absence of intervention.

The process begins with exposure to the causative agent capable of causing disease. Without medical intervention, the process ends with recovery, disability, or death. Most diseases have a characteristic natural history, although the time frame and specific manifestations of disease may vary from individual to individual. The usual course of a disease may be halted at any point in the progression by preventive and therapeutic measures, host factors, and other influences. The stages in the natural history of disease are shown in Figure 2.1.



The natural history of tuberculosis infection is illustrated in Figure 2.2. Acquisition of *M. tuberculosis* infection occurs when a susceptible person is exposed to droplet nuclei containing viable organisms that have been aerosolized from an infectious source, usually a patient with untreated pulmonary tuberculosis. A variety of non-immunologic host defenses are involved in preventing acid fast bacilli from establishing a productive infection in the lungs, including upper airway filtering mechanisms, impaction of organisms in the large and small airways, clearance of organisms by the mucociliary escalator, and killing of organisms by macrophages in the alveoli. Even after close contact with an infectious case, only about 30 percent of susceptible contacts acquire infection, as determined by a delayed-type hypersensitivity response to purified protein derivative. After the initial infection, intracellular replication of bacilli occurs, and dissemination of organisms may result from lymphatic and hematogenous routes. For the most part, however, the initial infection is clinically silent and there are no symptoms. In a minority of infected people, the initial infection may progress to active clinical disease within two years, defined as progressive primary tuberculosis. Risk factors for primary disease include immunosuppression (especially HIV infection), extremes of age, or a large inoculation. For the majority of infected persons, however, tuberculosis remains clinically and microbiologically latent for many years. Reactivation tuberculosis occurs years after the initial infection in a small proportion of patients, as well. Risk factors for reactivation tuberculosis disease include older age, immunosuppression (especially HIV), diabetes, renal insufficiency, intravenous drug use, malnutrition, and gastric or ileal bypass surgery. As depicted in the figure below, however, the vast majority of persons infected with *M. tuberculosis* never develop clinical disease.

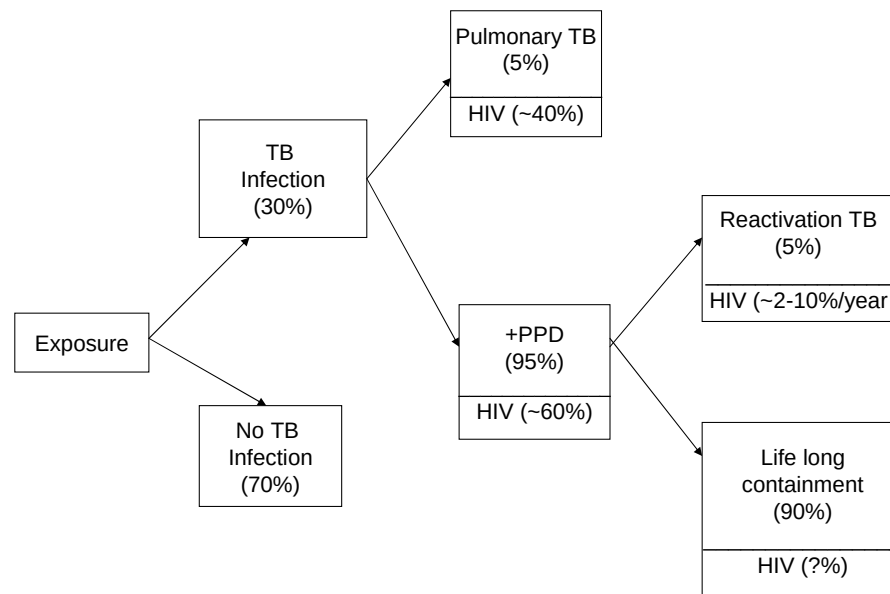


Figure 2.2 Natural History of Tuberculosis

C. Components of Infectious Disease Process

Infectious diseases result from the interaction of infectious **agent**, susceptible **host**/reservoir and **environment** that brings the host and the agent together.

Agent: Agent is an infectious micro-organism- virus, bacteria, parasite, or other microbe.

Host: Host factors influence individual's exposure, susceptibility or response to a causative agent. For example- age, sex, race, socioeconomic status, and behaviours (smoking, drug abuse, lifestyle, sexual practices and contraception, eating habits) affect exposure.

Environment: Environmental factors are extrinsic factors which affect the agent and the opportunity for exposure. Physical factors such as geology, climate, and physical surrounding (e.g., maternal waiting home, hospital); biologic factors such as insects that transmit the agent; and socioeconomic factors such as crowding, sanitation, and the availability of health services.

D. Causal Concepts of Disease

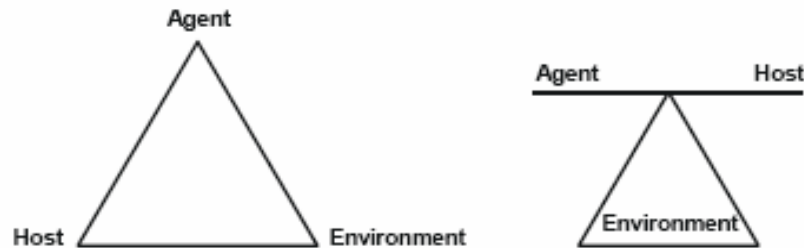
Not all associations between exposure and disease are causal. A cause of a disease can be defined as a factor (characteristic, behaviour, event, etc.) that influences the occurrence of disease. If disease does not develop without the factor being present, then we term the causative factor "necessary". If the disease always results from the factor, then we term the causative factor "sufficient".

Example: Tubercle bacillus is a necessary factor for tuberculosis.

Rabies virus is sufficient for developing clinical rabies.

The epidemiologic triad or triangle is the traditional model of infectious disease causation. It has three components: an external agent, a susceptible host, and an environment that brings the host and agent together, as shown in the two diagrams in Figure 2.3.

Figure 2.3
EPIDEMIOLOGIC TRIANGLE AND TRIAD (BALANCE BEAM)



Examples of causes of disease by host, agent and environmental factors.

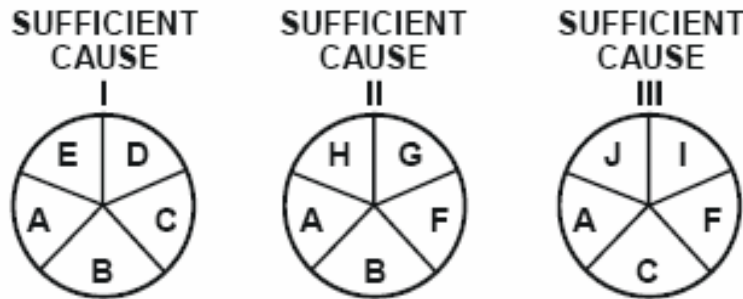
Host factors	Agent factors	Environmental factors
▪ Age ▪ Sex ▪ Previous disability ▪ Behaviour ▪ Genetic inheritance ▪ Height ▪ Weight	▪ Virulence of organisms ▪ Serotype of organisms ▪ Antibiotic resistance ▪ Cigarette-tar content ▪ Type of glass in motor car windscreen	▪ Home overcrowding ▪ Air pollution ▪ Workplace hygiene ▪ Weather ▪ Water composition ▪ Food contamination ▪ Animal contact

In recognition of the multi-factorial nature of most diseases such as heart disease and many cancers several other models have been proposed. Those models emphasize that there is no single cause, causes of disease are interacting, disentangling the cause is highly impossible, and causality may be two ways (reverse causality).

Causal pie is one of the models that take into account multiple factors which are important in causation of disease. In the causal pie model, the factors are represented by pieces of the pie called component causes, as shown in Figure 2.4.

In disorders with multi-factorial causation often no specific causes are known, many factors appear to be important, and mechanisms of causation are not apparent. Sometime our understanding of the disease could be too rudimentary to permit the use of simplified models. Models such as the Wheel of causation and spider's web are attempts to portray complex causation interactions. The purpose of the models is to simplify reality and make easier to grasp the essence of the issue. Narrow causal thinking based on single causes can be misleading; pointing to premature believing that a problem is solved and can seriously distort public health action.

Figure 2.4
Rothman's Causal Pies: Conceptual Scheme for Disease Causation*



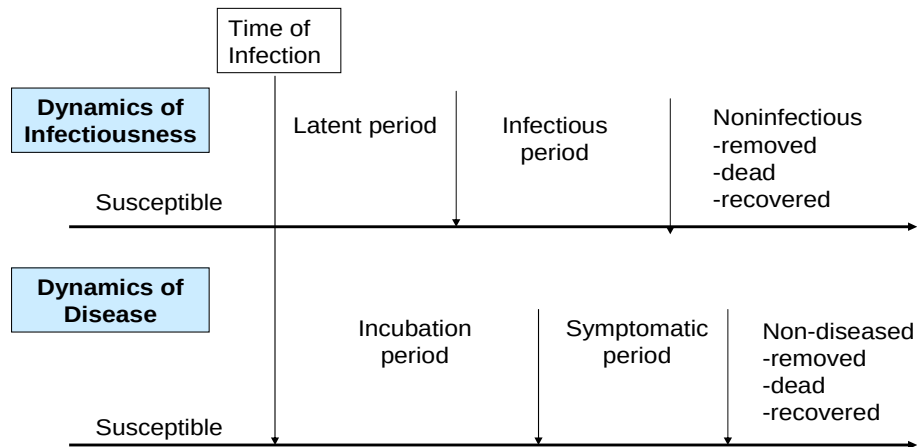
* All factors (component causes) together form the sufficient cause while component cause A constitutes the necessary cause.

E. Time lines of Infection

The time lines of infection begin with the successful infection of the susceptible host by infectious agent. The time line of infectiousness includes the latent period (the time interval from infection to development of infectiousness) and the period of infectiousness of the host, during which time the host could infect another susceptible host. The host becomes non-infectious either by recover from the infection or by death. The host can also become non-infectious while still alive and still harbouring the parasite.

The time line of disease within the host includes the incubation period (the time from infection to development of symptoms of the disease), and the symptomatic period. The probability of developing symptoms or disease after becoming infected is referred as pathogenicity. The host eventually becomes asymptomatic either by recovering from the symptoms or by death. Carrier state develops when the person become asymptomatic but remains infectious. An inapparent or silent infection is a successful infection that does not develop detected symptoms, they can be infectious.

Figure 2.5. Time lines for Infection and Disease.



F. Transmission Probability

The transmission probability is the probability that, given that there is contact between an infective source and a susceptible host, successful transfer of the infective agent will occur so that susceptible host become infected. The transmission probability depends on characteristics of infective source (infected person, mosquito vector, and contaminated inanimate object), the parasite, the susceptible host, and the type and definition of contact. The mode of transmission of an infective agent determines what types of contact are potentially infectious. For example; being in the same school on one day with someone with culture-proven whooping cough or living in the same house during the period of presumed infectiousness of the person with whooping cough; and sexual act with a person infected with HIV or partnership with a person infected with HIV.

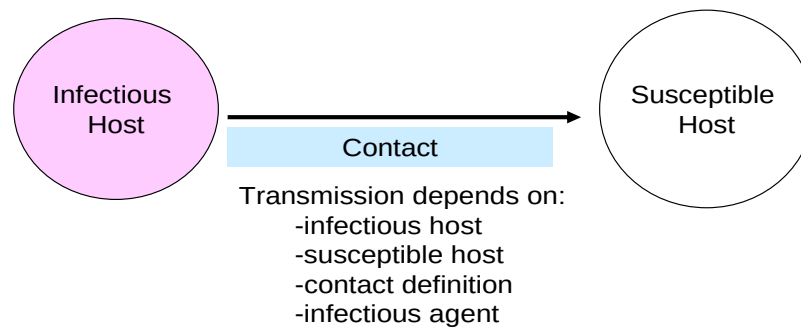


Figure 2.6 Disease Transmission from Infectious host to susceptible host.

G. Estimating the Transmission Probability (Measures of Transmission Probability)

There are two common ways of estimating transmission probability.

1. Secondary attack rate
2. Binomial Model

G. 1. Secondary Attack Rate

A secondary attack rate is a measure of the frequency of new cases of a disease among the contacts of known cases. It requires identifying infectious persons and susceptible people who make contact with them by some definition of contact, such as being in the same classroom. SAR is a proportion, not a rate. It is often defined for exposure to an infective within some small population unit, such as household, classroom, prison, or school bus. The formula is as follows:

$$\text{SAR} = \frac{\text{Number of persons exposed (have contact with a known case) who develop disease}}{\text{Total number of susceptible exposed person}} \times 10^n$$

The first step in assessing the SAR is to define for disease under study the time interval after the index case that would include secondary cases (cases with onset of symptoms between minimum and maximum incubation period); a case with recorded onset time less than one minimum incubation period after that of the index case is called co-primary case, it is not presumably infected by the index case (Figure 2.7). The data required for estimating secondary attack rate

are:

- the time of onset of disease for each case in the household;
- knowledge of who is susceptible;
- estimates/assumptions about minimum and maximum incubation periods;
- the latent period; and
- the maximum time that a person remains infectious; sometimes it can assumed that the onset of symptoms coincides with the onset of infectiousness and that there are no inapparent cases.

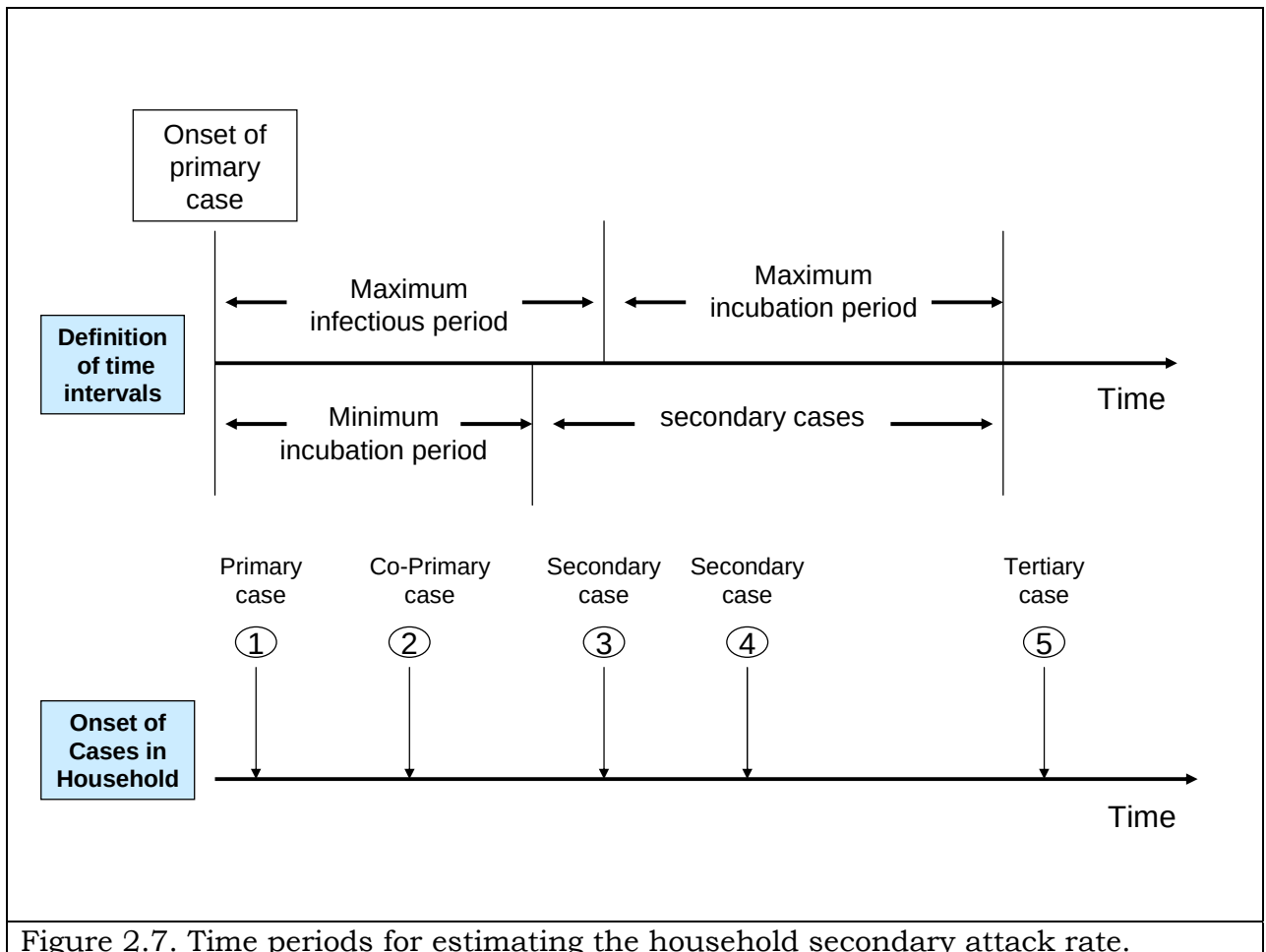


Figure 2.7. Time periods for estimating the household secondary attack rate.

G.2. Binomial Models of Transmission Probabilities

This model is often used when susceptible individuals are exposed to more than one potentially infectious case. The following notations and formulas are used for this model:

P	The probability of transmission during a contact between a susceptible and infectious person
$Q=1-p$	The probability of the susceptible person's escaping infection during the contact
N	Number of contacts with an infective or with different infective individuals
$q^n=(1-p)^n$	The probability of escaping infection from all n potentially infective contacts
$1-q^n=1-(1-p)^n$	The probability of being infected after n contacts (the probability of not escaping infection from all n contacts)
$P^\wedge$	The maximum likelihood estimate if the transmission probability under the binomial model is: $\frac{\text{Number of susceptible who become infected}}{\text{Total number of contacts with infectives}}$ Note: -The numerator is the same as for secondary attack rate (SAR). - The denominator in here is total number of potentially infectious contacts that susceptible individuals make, while in SAR each susceptible person had just one potentially infectious contact with infective case. - The two formulas would be the same if everyone in the binomial model made just one potentially infectious contact.

H. Basic Reproductive Number (R_0)

Basic reproductive number is defined as the expected number of new infectious secondary cases (does not include secondary cases who do not become infectious) that one infectious host will produce during the period of infectiousness in a large population that is completely susceptible; it does not include the new cases produced by the secondary cases or further down the chain. For an epidemic to occur in a susceptible population R_0 must be >1 . If R_0 is <1 , an average case will not reproduce itself, so an epidemic will not spread. Since R_0 is an average, even when R_0 is <1 , it is possible that a particular case will produce more than one infective case producing a small cluster of cases that is unlikely to be a self-sustaining outbreak.

As R_0 is the number of new infectious cases per infectious case it is a dimensionless quantity, thus one cannot conclude about the time frame of an epidemic based on R_0 .

c	rate of contact			
d	duration of infectiousness			
P	the transmission probability per potentially infective contact			
cd	the average number of contacts made by an infective case during the infectious period			
$R_0 =$	number of contacts per unit time	x transmission probability per contact	x duration of infectiousness	=cpd

R_0 assumes that all contacts by the infective case are with susceptible individuals. In reality, there are often people who are already immune to an infective agent. Under these circumstances, the expected number of new cases produced by an infective case is less than R_0 and is called the effective reproductive number, which is denoted by R . If x is the proportion of susceptible population, R is the product of the basic reproductive number and the proportion of susceptible contacts.

$$R = R_0 x$$

I. Chain of Infection

Infection implies that the agent has achieved entry and begun to develop or multiply, whether or not the process leads to disease. A model used to understand the infection process is called the chain of infection (Figure 2.8). Each link must be present and in sequential order for an infection to occur. The links are: infectious agent, reservoir, portal of exit from the reservoir, mode of transmission, and portal of entry into a susceptible host. Understanding the characteristics of each link provides with methods to prevent the spread of infection. Sometimes the chain of infection is referred as the transmission cycle.

Components of Chain of Infection

1. Causative Agent
2. Reservoir host
3. Portal of exit
4. Mode of transmission
5. Portal of entry
6. Susceptible **host**.

Figure 4
CHAIN OF INFECTION

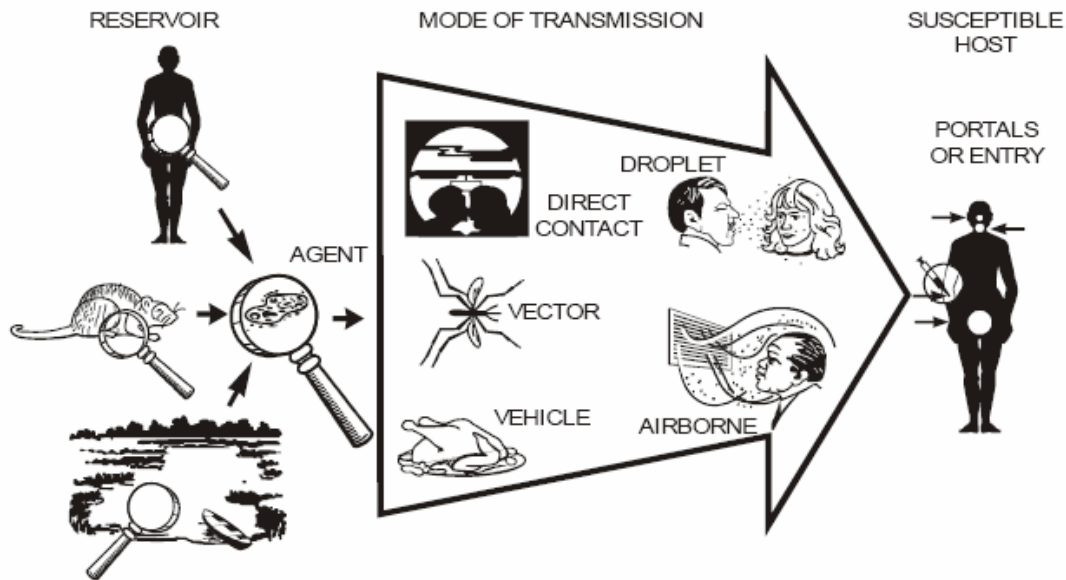


Figure 2.8. Chain of infection

The reservoir of an agent is the habitat in which an infectious agent normally lives, grows, and multiplies. Agents with a human reservoir include measles, mumps, and most respiratory pathogens. Human reservoirs may be persons with symptomatic illness, or carriers. A carrier is a person without apparent disease who is nonetheless capable of transmitting the agent to others (Figure 2.9). The importance of carriers in the transmission of disease depends on their: 1) number, 2) detectability, 3) mobility, and 4) chronicity.

Carriers may be:

- Asymptomatic carriers (transmitting infection without ever showing signs of the disease),
- Incubatory carriers (transmitting infection by shedding the agent before the onset of clinical manifestations), or
- Convalescent carriers (transmitting infection after the time of recovery from the disease).
- Chronic carriers shed the agent for a long period of time, or even indefinitely.

The chain of infection may be interrupted if the agent does not find a susceptible host. This may occur if a high proportion of individuals in a population is resistant to the agent. Through such herd immunity, immune persons limit the spread of the infection to the relatively few who are susceptible by reducing the probability of contact between infected and susceptible persons. Herd immunity operates best when there is: 1) a single

reservoir, 2) direct transmission, 3) total immunity, 4) no shedding of the agent by immune hosts, 5) a uniform distribution of immunes, and 6) no overcrowding.

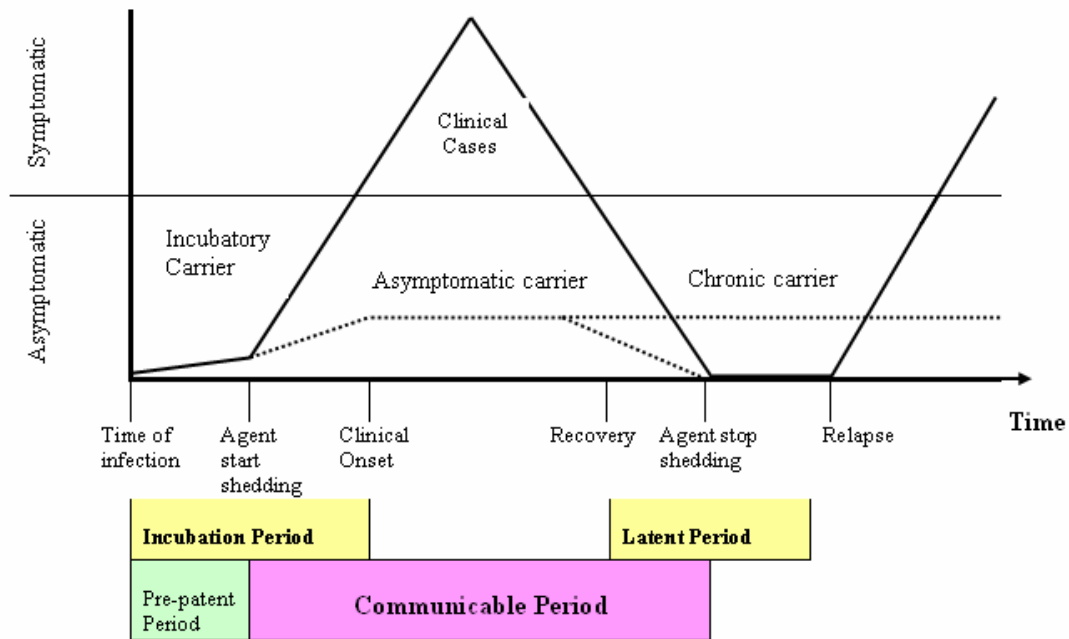


Figure 2.9. Time Course of a Disease in Relation to Its Clinical Expression and Communicability

J. Modes of Transmission of Infectious Agents

The mechanism by which an infective agent exists from a reservoir host and enters into a susceptible host is referred as mode of transmission. There are two major modes:

J.1. Direct Transmission- immediate transfer of the agent from a reservoir to a susceptible host by direct contact or droplet spread.

Example:

- Touching
- Kissing
- Sexual intercourse
- Blood transfusion
- Trans-placental

J.2. Indirect Transmission- an agent is carried from reservoir to a susceptible host by suspended air particles or by animate (vector-mosquitoes, fleas, ticks...) or inanimate (vehicle-food, water, biologic products, fomites) intermediaries.

Example:

- Vehicle-borne: food, water, towels, ...
- Vector-borne: insect animals, ...
- Airborne: dust, droplets
- Parenteral injections

K. Levels of Disease Prevention

Disease prevention means to interrupt or slow the progression of disease. Therefore, the aim is to push back the level of detection and intervention to the precursors and risk factors of disease. Fluctuation in patterns of morbidity and mortality over time in countries and the observation that migrants slowly develop the patterns of disease of host populations indicate that causes of disease are preventable. Hence, epidemiology plays a central role in disease prevention by identifying those modifiable causes. The levels of prevention in relation to the stage of the disease process are shown in Table 2.1.

Table 2.1. Levels of prevention in relation to the stage of the disease.

Level of Prevention	Stage of disease	Target
Primordial	Existence of underlying condition leading to causation The aim is to avoid the emergence and establishment of the social, economic, and cultural patterns of living that are known to contribute to an elevated risk of disease. Example: smoking, environmental pollution	Total population and selected groups
Primary	Specific causal factors exist The causative agent exists but the aim is to prevent the development of disease. Example: immunization Measles, polio	Total population, selected groups and health individuals
Secondary	Early stage of disease The aim is to cure patients and prevent the development of advanced disease. Example: Early detection & treatment of cases of tuberculosis & STD	Patients
Tertiary	Late stage of disease(treatment & rehabilitation) The aim is to prevent severe disability and death. Example: Leprosy	Patients

L. Levels of Disease Occurrence

Diseases occur in a community at different levels at a particular point in time. Some diseases are usually present in a community at a certain predictable level, this is called the expected level, but at times disease may occur in excess of what is expected (Figure 2.10).

1. Expected levels

- a) **Endemic:** a persistent level of low to moderate occurrence
- b) **Hyper-endemic:** a persistently high level of occurrence
- c) **Sporadic:** occasional cases occurring at irregular intervals

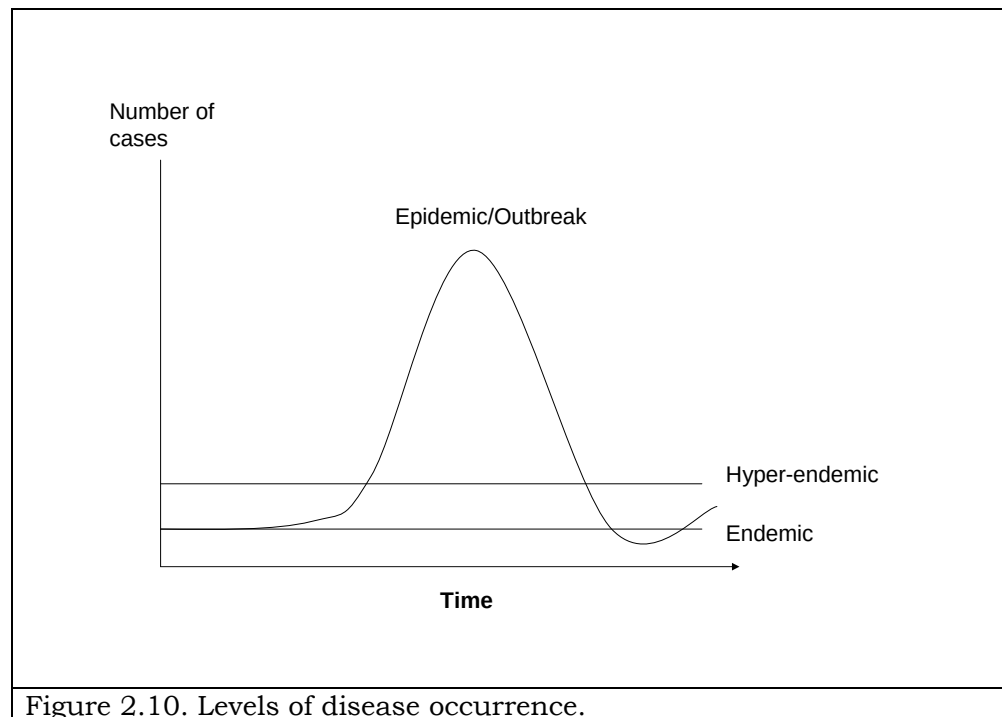


Figure 2.10. Levels of disease occurrence.

2. Excess of what is expected

- a) **Epidemic:** occurrence of disease in excess of what is expected in a limited period.
- b) **Outbreak:** same as epidemic, often used by public health officials because it is less provocative to the public.
- c) **Pandemic:** an epidemic spread over several countries or continents, affecting a large number of people.

* Disease Clustering: this is a rather confusing terminology and its use must be carefully understood.

- Disease cluster is defined as an aggregation of relatively rare events or diseases in time and/or place.
- The terms clusters and clustering should not be used in the context of common diseases since clustering is inevitable due to chance alone, or for infectious diseases that spread from person-

- to-person.
- A disease cluster is a mini-epidemic of a rare event in which occurrence of the disease is clearly in excess of that expected. Clusters may provide useful clues to public health action but often they are difficult to handle because of small number. Clusters are special instances of disease variation in a locality or over a short time period.

M. Disease Classification

Disease is often classified according to: 1) its time course, or 2) its cause. The time course classifies disease as acute (characterized by a rapid onset and short duration) or chronic (characterized by a prolonged duration). A chronic disease may have both acute and chronic manifestations. The cause of a disease may be classified as infectious (caused by living organisms which are transmissible) or non-infectious.

The outcomes of exposure to an infectious agent (see figure 2.11) are referred as:

Infectivity: the proportion of exposed persons who become infected.

Pathogenicity: the proportion of infected persons who develop clinical disease.

Virulence: the proportion of persons with clinical disease who become severely ill or die.

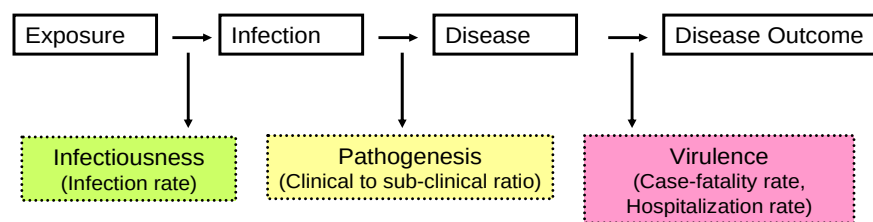


Figure 2.11. Outcomes at Each Stage of Infection

N. Variation in Severity of Illness

The infectious process has a wide spectrum of clinical effects which ranges from inapparent infection to severe clinical illness or death (Figure 2.12). The effect depends on the nature of the infectious agent and host susceptibility. Case fatality rate (CFR) is the measure of severity of illness.

$$* \text{ CFR} = \frac{\text{Number of deaths from a disease}}{\text{Number of clinical cases of that disease}}$$

Recognizing inapparent infections require the use of laboratory tests on seemingly healthy individuals. Information thus obtained are useful in planning public health interventions. A good example could be HIV testing to determine the potentials for the spread of the disease and to plan appropriate control strategies.

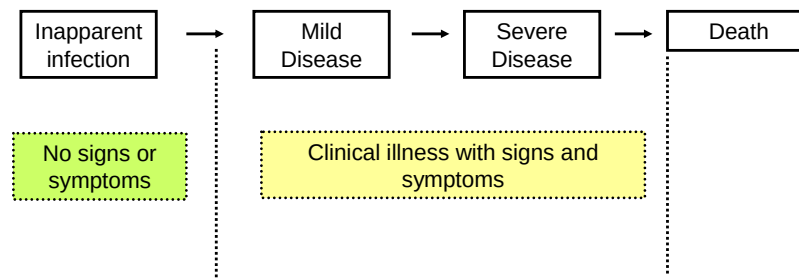


Figure 2.12. The Spectrum of Illness from Communicable Disease.

O. Spread of Disease through Person to Person transmission

Person to person transmission of an infectious agent is one of the main methods of disease spread in a community and is dependent on:

1. Generation time: This refers to the period between exposure/infection and the maximum communicability of the exposed host regardless of whether the disease is apparent or inapparent. In case of apparent infections generation time may be equivalent to incubation period.
2. Herd immunity: This refers to a community resistance to spread of an infectious agent as a result of immunity gained by high proportion of individual members of the community. Though it may not be important to achieve 100% immunity, successful breakage of the chain of infection can be achieved if the immunity is close to 100%.
3. Secondary attack rate: This is an important measure of spread of disease among contacts of an index case. It has great use in epidemic situations.

Secondary AR = $\frac{\text{New cases among contacts of index cases during the period}}{\text{Total number of contacts with the index cases}}$

* The index cases are excluded from both numerator and denominator.

* Index case: The case that brings a household or any other group (community) to the attention of the public health personnel.

Attack Rate(AR)

An attack rate is a variant of an incidence rate, applied to a narrowly defined population observed for a limited time, such as during an epidemic. It is usually expressed as a percent.

AR = $\frac{\text{New cases among the population during the specified period}}{\text{Population at risk at the beginning of the period}}$

III. OVERVIEW OF EPIDEMIOLOGIC STUDIES

The purpose of the broad category of epidemiological studies is given in Table 3.1. In this section students are expected to understand fully the distinction between these two broad categories.

Figure 3.1. Purpose of Epidemiological Studies

Descriptive	Analytic
Characterize disease occurrence by time, place and person.	Concerned with the search for causes and effects.
Generate testable Hypothesis as to the cause of disease.	Test hypothesis about association between exposure and outcome.

3. A. Descriptive Epidemiology

Descriptive epidemiology is a way of organizing data related to health and health related events by person (Who), place (Where) and time (When) in a population. Information organized as such is easy to communicate and provides information about:

- 1) the magnitude of the problem,
- 2) the populations at greatest risk of acquiring a particular disease, and
- 3) the possible cause(s) of the disease.

Time - Information organized by time easily shows the trend of the disease over time and establishes the usual occurrence of the disease in the population which is essential in identifying excess occurrence (epidemics). It can also be used to predict seasonal and secular (long-term) trends.

Place - This provides information on geographic distribution of the disease. Such information provides clue in identifying factors influencing the occurrence of the disease either in the host or environment.

Person - Describing disease occurrence by personal characteristics is important to identify some modifiable factors in order to prevent or control the disease. Person data include: the inherent characteristics of people (age, ethnic group, gender), their acquired characteristics (educational marital, immune, or nutritional status), their activities (occupation, leisure activities, use of alcohol, tobacco, or medications), or the conditions in which they live (socioeconomic status, access to health care).

3.B. Analytic Epidemiology

Analytic epidemiology uses **comparison groups** to determine whether the characteristics of those with a given health condition are alike or different from that expected. When persons with a particular characteristic are more likely than those without the characteristic to develop a certain health

problem, we say that the characteristic is **associated with** that health problem. Thus analytic epidemiology is concerned with the search for causes and effects, or the *why* and *how*. We use analytic epidemiology to quantify the association between exposures and outcomes and to test hypotheses about causal relationships.

The principles of analytic epidemiology are applicable to all types of disease, whether acute or chronic, infectious or non-infectious (even to states of health, behaviours, and phenomena). They are most often associated, however, with studies of chronic disease. This reflects the difference between chronic and acute disease in terms of: 1) duration of latency (short for acute, long or variable (or obscure) for chronic disease, 2) magnitude of incidence (usually high for acute disease, low for chronic), and complexity of causation (single causes are common for acute disease, multi-factorial aetiologies for chronic). Because of these characteristics, chronic diseases more often require relatively elaborate studies to establish causes, while acute disease can often be linked to its cause through simple descriptive or short-term cross-sectional studies.

Analytic epidemiology uses two categories of studies to understand causes and effects: 1) experimental studies and 2) observational studies. In an **experimental study**, we determine the exposure status for each individual (clinical trial) or community (community trial); we then follow the individuals or communities to detect the effects of the exposure. In an **observational study**, which is more common, we simply observe the exposure and outcome status of each study participant.

Two types of observational studies are the cohort study and the case-control study. A **cohort study** is similar in concept to the experimental study, except that we observe the exposure status rather than determining it. Cohort studies categorize subjects on the basis of their exposure and observe the frequency of disease occurrence. **Case-control studies** enrol a group of people with disease ("cases") and a group without disease ("controls") and compare their patterns of previous exposures to risk factors.

IV. MEASUREMENT IN EPIDEMIOLOGY

Measurement in Epidemiology

1. Measures of disease occurrence
 - Prevalence
 - Incidence
2. Standardization
3. Measures of association
 - Rate ratio
 - Etiologic fraction
4. Variations in disease occurrence and associations

4. A. Epidemiologic Variables

Variation in disease pattern is the foundation of epidemiology. Anything which varies and has different values is known as variable. In epidemiology there are two types of variables: exposure and outcome. The common epidemiological variables such as age, sex, economic status, social class, occupation, area of residence, religion and ethnicity are all powerful ways of showing variations in broad range of diseases and health status. However, most of these variables are markers for complex, underlying phenomena of interest which cannot be measured directly and easily. For example, sex may act as a proxy for genetic, hormonal, psychology or social status in different studies.

A good epidemiological variable should have the following attributes (see the example given for age in Table 4.1):

- have an impact on health in individuals and populations;
- be measurable accurately; differentiate populations in their experience of disease or health;
- differentiate populations in some underlying characteristics relevant to health, e.g. income, childhood circumstance, hormonal status, genetic inheritance, or behaviour relevant to health;
- generate testable aetiological hypotheses, and/or
 - help to develop health policy, and/or
 - help to plan and deliver health care, and/or
 - help prevent and control disease.

Table 4.1 Assessing Age as an Epidemiological Variable.

Criteria for good epidemiological variable	Criteria in relation to age
Impact on health in individuals and populations	Age is a powerful influence on health
Be measurable accurately	In most populations age is measurable to the day, but in some it has to be guessed
Differentiate populations in their experience of disease or health	Huge differences by age are seen for virtually every disease, health problem, and for factors which cause health problems
Differentiate populations in some underlying characteristics relevant to health	Differences in disease patterns in different age groups reflect a rich mix of environmental factors and may also reflect population changes in genetic factors, particularly in populations where migration has been high
Differentiate populations in some underlying characteristics relevant to health	It is hard to test hypotheses because there are so many underlying differences between populations of different age
Help to develop health policy	Age differences in disease patterns profoundly affect health policy
Help to plan and deliver health care	Knowing the age structure of a population is critical to good decision making
Help prevent and control disease	By understanding the age at which diseases start, preventive and control programmes can be targeted at appropriate age groups.

4.B. Measures of Disease Occurrence

The number of cases in a given community can give more epidemiologic sense if they are related to the size of the population. Such tie of the number of cases with the population size can be determined by calculating ratios, proportions, and rates. These measures provide useful information about the probability of occurrence of health events, population at a higher risk of acquiring the disease. They are also important in designing appropriate public health interventions.

Ratio: the value of x and y may be completely independent, or x may be included in y .

Example: Male: Female (male to female ratio)

Proportion: is a ratio (expressed as a percent) in which x is included in y .

Example: Male/Both sexes (proportion of male in a community)

Rate: measures the occurrence of an event in a population over time. The time component is important in the definition. Rates are often proportions. Rates must: 1) include persons in the denominator who reflect the population from which the cases in the numerator arose; 2) include counts in the numerator which are for the same time period as those from the denominator; and, 3) include only persons in the denominator who are "at risk" for the event.

Example: Measles cases in under five in 1995
Under five children in 1995.

When we call a measure a **ratio** we usually mean a non-proportional ratio. When we call a measure a **proportion**, we usually mean a proportional ratio that doesn't measure an event over time. When we use the term **rate**, we frequently refer to a proportional ratio that does measure an event in a population over time. The following table depicts the common uses of the three measures. Table 4.2 displays the common use of the measures of disease occurrence.

Table 4.2. Common uses of measures of disease occurrence.

	As measure of disease occurrence	As measure of comparison	As measure of Impact of intervention
Ratio	-	Rate Ratio(OR, RR)	Rate Ratio
Proportion	Prevalence	-	Etiologic fraction
Rate	Incidence	-	-

The measures used to show frequency of events related to morbidity, mortality, and natality are described in Table 4.3. Students are advised to look the formulas for each of the specific measures in Table 4.3.

Table 4.3. Measures Described by Type of Event

Event	RATIOS	PROPORTIONS	RATES
Morbidity (Disease)	Relative risk Odds Ratio	Attributable Proportion Point Prevalence	Incidence Rate Attack Rate Period Prevalence
Mortality (Death)	Death-to-Case Ratio Maternal Mortality Rate Proportionate Mortality Ratio Postneonatal Mortality Rate	Proportionate Mortality Case-Fatality Rate	Crude Mortality Rate Cause-Specific Mortality Age-Specific Mortality Sex-Specific Mortality Race-Specific Mortality Age-Adjusted Mortality Neonatal Mortality Infant Mortality Years of Potential Life Lost
Natality (Birth)		Low Birth Weight	Crude Birth Rate Crude Fertility Rate Rate of Natural Increase

4.C. Common Measures of Disease Frequency

The frequency of health related events are measured by risk, prevalence and incidence rate.

Risk (cumulative incidence):

- Likelihood that an individual will contract a disease.
- The proportion of unaffected individuals who, on average, will contract the disease of interest over a specified period of time.

$$\text{Risk} = \frac{\text{New cases occurring during a given time period}}{\text{Population at risk during the same period}}$$

Prevalence:

- The amount of disease that is present already in a population.
- Indicates the number of existing cases in a population.

$$\text{Prevalence} = \frac{\text{all new and pre-existing cases during a given time period}}{\text{population during the same time period}}$$

Incidence:

- Measures the rapidity with which newly diagnosed patients develop over time.
- Most common way of measuring and comparing the frequency of disease in populations.
- The period of time for the rate must be specified.

$$\text{Incidence Rate} = \frac{\text{Number of new cases during observation period}}{\text{Person-time observed}}$$

4.D. Standardization

Crude rates apply to the total population of a given area. Specific rates apply to specific subgroups in the population (such as by age, sex, or occupation) or specific diseases. When numerator and denominator are precisely available age and sex specific rates can be calculated and compared between times, places, and population groups. These rates provide the undistorted view of the disease patterns and should be presented wherever possible. When age and sex specific rates could be imprecise due to inadequate sample size, when trying to present a summary overall rate for planning and intervention purposes, making comparisons of population in comparative research over all crude (actual) rates may mislead. Adjusted rates and age-specific rates are often used to permit comparison of mortality rates in populations which differ in age and sex structure; when age and sex are confounding the overall rate. Mortality rates computed with adjustment techniques are called standardized or adjusted rates. Often standardization are made for age and sex, but not limited to them.

Summarizing age-specific rates into one age-adjusted figure may have disadvantages such as:

- They are not true population based rates and as such do not accurately measure the health status of a population. The health care needs developed based on adjusted estimates are wrong.
- By summarizing into one figure there is a loss of important information when differences are not consistent across age group or sex.

Two different standardization techniques are used to adjust for the effects of the differing age structures and make overall comparisons possible.

1. Direct Standardization: this technique applies the age-specific rates from the study population to a standard population structure. The choice of the standard population structure has effect on the standardized estimate; when dealing with age-specific rates that consistently increasing with age the use of young population structure gives lower estimate compared to the estimate obtained using older population structure.

The Table below shows that although the age-specific rates are the same in all three populations the crude varies remarkably due to the difference in population size. Therefore, the overall crude rate is confounded by age. In order to nullify the effect of the differing age structure a direct method of standardization is illustrated in Table 4.4 using two population structures; young and old population.

Table 4.4. Age-specific and crude rates in three hypothetical populations.

Age group	Population size	Cases	Rates (%)
Population A			
15-29	2000	100	5
30-44	2000	200	10
45-59	2000	300	15
Overall crude rate	6000	600	10
Population B			
15-29	1000	50	5
30-44	3000	300	10
45-59	6000	900	15
Overall crude rate	10000	1250	12.5
Population C			
15-29	10000	500	5
30-44	2000	200	10
45-59	400	60	15
Overall crude rate	12400	760	6.1
Population D			
15-29	3000	300	10
30-44	3000	450	15
45-59	3000	600	20
Overall crude rate	9000	1350	15

The age-specific rates show that the disease rates are identical in three of the populations and rise with age in all populations. Population D has the highest crude rate because it has the highest age-specific rates. Population B has the highest crude rate among the three populations that have identical age-specific rates because it has a comparatively older population.

As shown in Table 4.5 direct standardization give the same number of expected cases and standardize rates for the populations (A,B&C) with identical age-specific rates whether one uses young or old population although the overall rate is higher when using the older population. Whereas the overall standardized rates for population D are different from the others because the age-specific rates are different. These standardized figures are useful for comparison purposes but since they are not real values they may mislead health service planning.

Table 4.5. Crude rates standardized with direct method showing the effect of young and old standard populations.

Age group	Population size	Expected Cases: by applying age-specific rates to standard population			
		Population A	Population B	Population C	Population D
Young Standard Population					
15-29	6000	300	300	300	600
30-44	3000	300	300	300	450
45-59	1000	150	150	150	200
Overall crude rate	10000	750	750	750	1250
Overall Standardized rate		$750/10000$	$750/10000$	$750/10000$	$1250/10000$
		=7.5%	= 7.5%	=7.5%	=12.5%
Older standard Population					
15-29	1000	50	50	50	100
30-44	3000	300	300	300	450
45-59	6000	900	900	900	1200
Overall crude rate	10000	1250	1250	1250	1750
Overall Standardized rate		$1250/10000$	$1250/10000$	$1250/10000$	$1750/10000$
		=12.5%	= 12.5%	=12.5%	=17.5%

- Indirect standardization: Calculation of a directly standardized rate relies on age-specific mortality rates being available for each population to be compared. For many countries these may not be obtainable, or data for strata may be based on small numbers. For these reasons, a more commonly used method of standardization called indirect standardization is used. In indirect standardization the standard (reference) population supplies disease rates, not population structure. This technique helps answer the question: how many cases would have occurred if the study population had the same rates as the standard population? In the example below the expected number of cases are different from what is observed (the actual number of cases) because the standard rates are weighted differentially by the different population structures. Note also the difference in the number of expected cases when using different standard rates (in the Tables 4.6 A&B) high and low rates are illustrated).

The Standard Mortality/Morbidity Ratio (SMR) is a summary output of the indirect adjustment which is weighted (biased) in relation to the age and sex structure of the population under study. Thus, comparing SMRs from several study populations is impossible. Only comparisons between the study population and the standard population are valid. Where the aim is to compare several populations either the specific rates or rates adjusted by the direct method should be examined.

Table 4.6.A Standardization with the indirect method.

Standard Population with High Rates								
Age Group	Population n	Cases	Rates (%)					
15-29	80000	8000	10					
30-44	100000	15000	15					
45-59	120000	24000	20					
Total	300000	47000	15.7					
	Population A		Population B		Population C		Population D	
	Population	Expected Cases	Population	Expected Cases	Population	Expected Cases	Population	Expected Cases
15-29	2000	200	1000	100	10000	1000	3000	300
30-44	2000	300	3000	450	2000	300	3000	450
45-59	2000	400	6000	1200	400	80	3000	600
Total	6000	900	10000	1750	12400	1380	9000	1350
Overall rate (Standardized)	900/6000 = 15%		1750/10000 = 7.5%		1380/12400 = 11.1%		1350/9000 =15%	
Observed/expected (Standardized Mortality Ratio, SMR)	600/900 =66%		1250/1750 =71%		760/1380 =55.1%		1350/1350= 100%	

Table 4.6.B Standardization with the indirect method.

Standard Population with Low Rates								
Age Group	Population n	Cases	Rates (%)					
15-29	80000	4000	5					
30-44	100000	7500	7.5					
45-59	120000	12000	10					
Total	300000	23500	7.7					
	Population A		Population B		Population C		Population D	
	Population	Expected Cases	Population	Expected Cases	Population	Expected Cases	Population	Expected Cases
15-29	2000	100	1000	50	10000	500	3000	150
30-44	2000	150	3000	225	2000	150	3000	225
45-59	2000	200	6000	600	400	40	3000	300
Total	6000	450	10000	875	12400	690	9000	675
Overall rate (Standardized)	450/6000 = 7.5%		875/10000 = 8.8%		690/12400 = 5.6%		675/9000 =7.5%	
Observed/expected (Standardized Mortality Ratio, SMR)	600/450 =133%		1250/875 =143%		760/690 =110%		1350/675= 200%	

4.E. Measures of Association

A. Rate Ratio

Measures of association between risk factors and disease are often calculated from data presented in a two by two table (Table 4.7).

Table 4.7. TWO-BY-TWO TABLE SHOWING ASSOCIATION

EXPOSURE	DISEASE	
	YES (+)	NO (-)
YES (+)	A	B
NO (-)	C	D

By convention, the capital letters (A,B,C,D) designate study populations (e.g., in cohort studies) defined by risk exposure and disease occurrence. The small letters (a,b,c,d) represent samples of populations (e.g., in case-control studies), usually of unknown and different sampling frequencies.

The relative risk or risk ratio compares the risk of some health-related event (often disease or death) in two groups, typically in persons exposed to the disease to those not exposed:

$$\frac{A}{A+B} \div \frac{C}{C+D}$$

Odds ratio, or cross-product ratio, is another measure of association which quantifies the relationship between an exposure and health outcome from a comparative study. The formula can be derived (to be the same as that for the relative risk) by dividing the odds that a case will have been exposed to the risk factor (a/c) by the odds that a control will have been exposed (b/d):

$$OddsRatio = \frac{\frac{a}{c}}{\frac{b}{d}} = \frac{ad}{bc}$$

When the health outcome is uncommon, the odds ratio provides a good approximation of the relative risk or risk ratio. The odds ratio is also useful in analysis of data from case-control studies, since the size of the control group is arbitrary and the true size of the population from which the cases come is usually not known. Under these circumstances, we cannot calculate incidence rates or the relative risk. The relative risk can, nonetheless, be

approximated by calculating the odds ratio particularly when the exposure in the control group represents the population from which cases are derived..

Since cases of disease in most chronic disease studies represent only a small fraction of exposed and unexposed populations, B is about equal to A+B and D to C+D. The formula can, under these circumstances, be simplified as follows:

$$\frac{\frac{A}{A+B}}{\frac{C}{C+D}} = \frac{\frac{A}{B}}{\frac{C}{D}} = \frac{AD}{BC}$$

A different procedure is used to calculate relative risk in case-control studies when controls are selected by matching. If matching is used in selecting controls, the matching (or pairing) should be retained for analytic purposes as shown in the following table of case-control pairs:

Table 4. TWO-BY-TWO TABLE FOR MATCHED CASE-CONTROL STUDY

Control Exposed to Risk Factor	Case Exposed to Risk Factor	
	YES	NO
YES	e	f
NO	g	h

In calculating relative risk, one need only consider discordant case-control pairs, represented by g (case exposed, control not exposed) and by f (control exposed, case not exposed).

B. Etiologic Fraction

This is an indirect method of estimating the effect of reducing or eliminating a causal factor. This will determine the relative benefit of public health interventions. In order to provide valid etiologic fraction information the following are needed:

- Valid evidence that the risk factor is a component of the causal pathway and not merely artifactually associated with the disease. Analytical studies usually provide such information. Causation must be judged using the criteria for establishing a causal association (see detail discussion in the relevant section of this lecture note).
- Knowledge of the frequency of the risk factor in the population. A rare risk factor has little effect on the incidence of the disease in the population.
- Estimate of the relative risk of developing the disease by the particular risk factor in the population (estimate of the relative risk). The impact of controlling a disease with relative risk of say 1.2 is much smaller than if the RR was 2.
- Understanding the nature and cost of the interventions required to reduce the prevalence of the risk factor in the population.
-

The attributable risk is the difference between the disease rate in exposed persons (or in the total population) and the rate in non-exposed:

$$\frac{A}{A+B} - \frac{C}{C+D}$$

The attributable proportion, also known as the attributable risk percent, is a measure of the public health impact of a causative factor. In calculating this measure, we assume that the occurrence of disease in a group not exposed to the factor under study represents the baseline or expected risk for that disease. Thus, we attribute any risk above that level in the exposed group to their exposure. It represents the expected reduction in disease if the exposure could be eliminated. The calculation is shown in the summary table.

Summary of Measures of Association

Attributable risk (AR) or Risk difference (RD) indicate how much of the risk is due to (or attributable to) the exposure. Quantify the excess risk in the exposed that can be attributable to the exposure by removing the risk of disease that could have occurred anyway due to other causes.

$$AR = \text{Risk in exposed} - \text{Risk in non-exposed}$$

$$=$$

Relative risk (RR): estimates the magnitude of the association between exposure and disease and indicates the likelihood of developing the disease in the exposed group relative to those who are not exposed.

$$RR = \frac{\text{Risk in exposed}}{\text{Risk in unexposed}}$$

Odds of exposure: is a simple ratio, not a proportion. Indicates odds of exposure relative to the disease status.

$$\begin{aligned} \text{Odds of exposure in diseased} &= a/c \quad \text{or} \quad a:c \\ \text{Odds of exposure in not-diseased} &= b/d \quad \text{or} \quad b:d \end{aligned}$$

Odds of disease: is a simple ratio, not a proportion. Indicates odds of diseased relative to the exposure status.

$$\begin{aligned} \text{Odds of disease in exposed} &= a/b \quad \text{or} \quad a:b \\ \text{Odds of disease in unexposed} &= c/d \quad \text{or} \quad c:d \end{aligned}$$

Odds Ratio (OR): is the chance of being exposed (or diseased) as opposed to not being exposed (or diseased). It is possible to calculate either exposure or disease odds ratio, which are exactly the same. The epidemiological thinking behind odds ratio is that if a disease is casually associated with an exposure, then the odds of exposure in the diseased group will be higher than the corresponding odds in the non-diseased group. It is also called cross product ratio.

$$\begin{aligned} \text{Exposure OR} &= a/c \div b/d = a/c \times d/b = ad/cb \\ \text{Disease OR} &= a/b \div c/d = a/b \times d/c = ad/bc \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Exposure OR} \\ \text{Disease OR} \end{aligned}} \right\} \begin{array}{l} \text{Results are} \\ \text{the same} \end{array}$$

Attributable Risk Percent (AR%) among exposed: estimate the proportion of disease among the exposed that is attributable to the exposure, or the proportion of the disease that could be prevented by eliminating the exposure.

$$AR\% = \frac{\text{Risk in the exposed} - \text{Risk in unexposed}}{\text{Risk in exposed}}$$

$$= \frac{RR - 1}{RR} \times 100 \quad \text{or} \quad OR - 1/OR \times 100$$

Population Attributable Risk (PAR) is the risk in total population minus risk in the non-exposed. Estimate the excess rate of disease in the total study population that is attributable to the exposure.

$$PAR = \text{Risk in population} - \text{Risk in unexposed}$$

Population Attributable Risk Percent (PAR%) Estimate the proportion of disease in the study population that is attributable to the exposure and thus could be eliminated if the exposure were eliminated.

$$PAR\% = \frac{\text{Risk in population} - \text{Risk in unexposed}}{\text{Risk in population}} \times 100$$

Possible Outcomes in studying the relationship between disease and exposure

1. No association between exposure and disease

Attributable risk = 0
 Relative risk/odds ratio = 1

2. Positive association between the exposure and the disease (i.e., more exposure, more disease)

Attributable risk > 0
 Relative risk/odds ratio > 1

3. Negative association between the exposure and the disease (i.e., more exposure, less disease)

Attributable risk < 0 (negative)
 Relative risk/odds ratio < 1 (a fraction)

* Association is dependent on your definition of exposure.

Example: Exposure is the sex of the Community Health Agent (CHA)

<u>CHA</u>	<u>Female</u>	<u>Male</u>
AR	>0	<0
RR/OR	>1	<1

====> The above summary indicates that there is a positive association with female CHA and negative association with male CHA.

Interpretation of Measures of Association

Rate Ratio: measures the strength of association between an exposure and disease and provide information that can be used to judge whether a valid observed association is likely to be causal.

Attributable Risk: measures the public health impact of an exposure, assuming that the association is one of cause and effect.

Relative and attributable risks of mortality from lung cancer and coronary heart disease among cigarette smokers in a cohort of British male physicians

	<u>Annual mortality rate per 100,000</u>	
	Lung cancer	CHD
Cigarette smokers	140	669
Nonsmokers	10	413
Relative risk	14.0	1.6
Attributable risk	130/10 ⁵ /year	256/10 ⁵ /year

The above study demonstrated a 14-fold increased death rate from lung cancer among smokers compared with non smokers. The relative risk of CHD mortality among current smokers compared with non smokers was 1.6. Thus, cigarette smoking is a much stronger risk factor for mortality from lung cancer than coronary heart disease. However, if smoking is causally related to both diseases, the elimination of cigarettes would prevent far more deaths among smokers from coronary heart disease than from lung cancer, as shown by the attributable risks of 256/100,000 and 130/100,000, respectively. The explanation for this is that while death from lung cancer is a relatively rare occurrence, accounting for only 10 deaths/100,000 population each year among non smokers, the annual death rate of coronary heart disease in that same group is 413/100,000. consequently, even a 60% increased risk of CHD mortality associated with cigarette smoking will affect a much larger number of people than a 14-fold increased risk of death from lung cancer. Thus, the potential public health impact of smoking cessation on mortality will be far greater for coronary heart disease than for lung cancer.

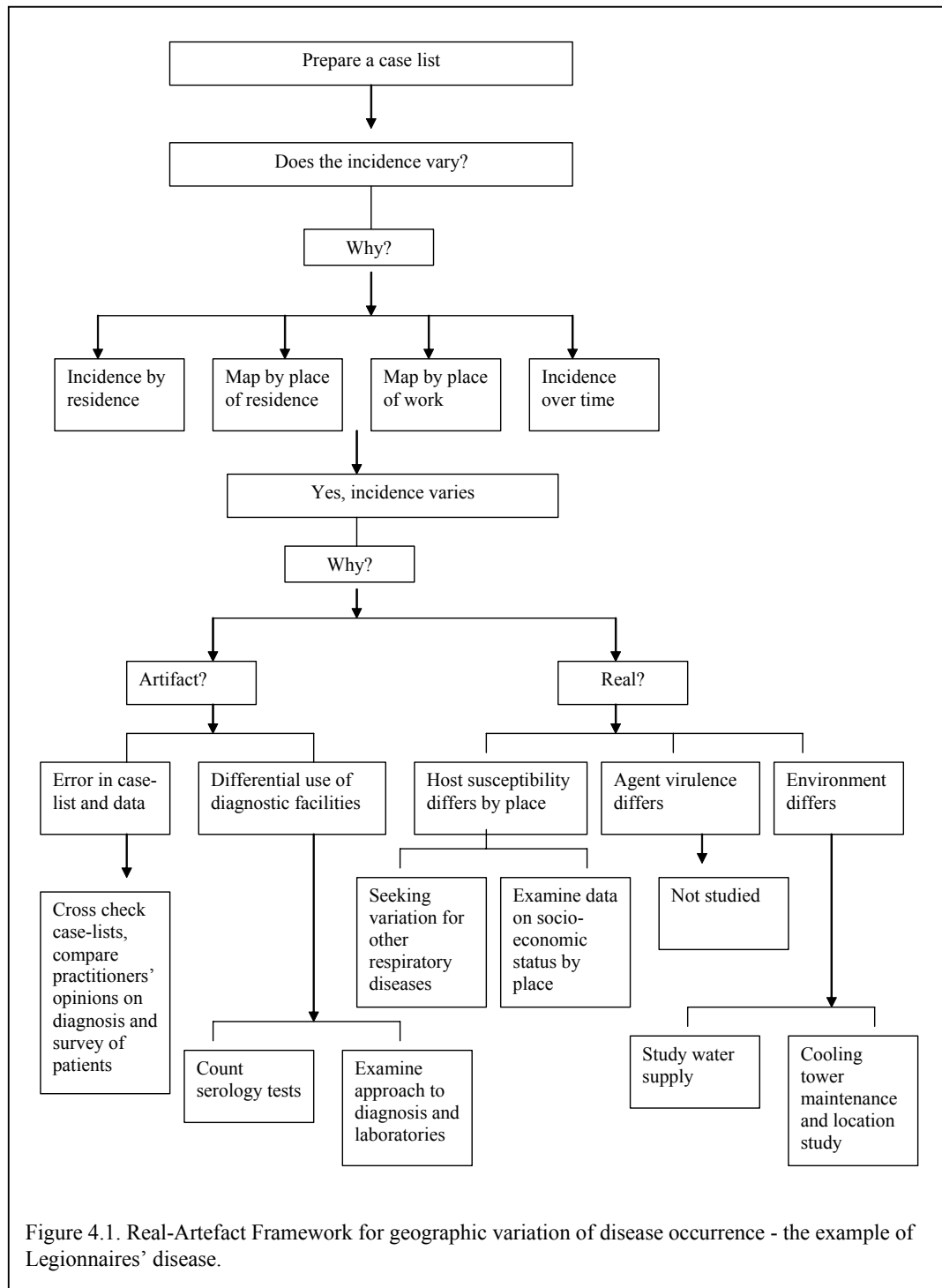
4.F. Variations in disease occurrence and associations

Changes in disease frequency could be due to two main reasons. The first reason is that changes are real (natural), and the second reason is that changes are due to mistakes/errors committed during diagnosing and counting (artefactual). As demonstration of disease variation is the basis for establishing epidemiological association it critical to examine whether variations are real or artefact. Table 4.9 gives some common reasons for real changes and sources of artefacts.

Table 4.9 Some common real and artefactual explanations for disease variation and associations.

Real explanations	Artefactual explanations
▪ Host factors: - o Genetic - o Behaviour: nutritional, social, medical ▪ Agent factors: - o Virulence - o Introduction of a new agent ▪ Environmental factors - o Housing: family size, homeless, prison - o Whether	▪ Chance: random fluctuation of cases over time. ▪ Errors of observation ▪ Change in size and structure of underlying population ▪ Health care seeking behaviour: alter the likelihood of being diagnosed and counted ▪ Diagnostic accuracy: changes in personal and diagnostic facilities ▪ Diagnostic method change ▪ Data collection method changes ▪ Changes in diagnostic code ▪ Change in analysis method ▪ Changes in the style of presentation of findings

The real-artefact framework provides a useful approach to systematically analyze variation of disease occurrence in a population. The process adapted to study environmentally acquired pneumonia (Legionnaires's disease) with no person-to-person spread is illustrated in Figure 4.1.



V. EPIDEMIOLOGICAL DESIGN STRATEGIES

Epidemiology is primarily concerned with the distribution and determinants of disease in human populations. In addressing these issues there are many approaches that are very difficult to clearly distinguish. The basic design strategies in epidemiologic research are categorized into two according to their focus of investigation (Table 5.1). Descriptive studies focus on the distribution of disease and analytic studies focus in elucidating the determinants of disease. However, there are a number of modifications to these basic designs in order to suit the specific purpose of studies. Knowing the fundamental differences in the basic strategies would permit understanding more complex and modified designs. Table 4.2 shows broad epidemiological design strategies.

Table 5.1 Types of Epidemiologic Design Strategies

DESCRIPTIVE	ANALYTIC
Dealing with population • Correlational or ecological	Observational studies • Case-control • Cohort
Dealing with individuals • Case report or series • Cross sectional survey	Intervention studies

Table 5.2 is a good way of illustrating that most epidemiological studies are of observational nature. This is one of the great advantages of epidemiology that is without altering the course of events deliberately a lot can be learned about health and disease in human population. The key questions in identifying the study designs are shown in Figure 5.1.

Table 5.2. Characteristics of the epidemiological designs.

Design	Descriptive/ Analytic	Retrospective/ Prospective	Observational/ Experimental	Beginning with disease/exposure	Presence of comparison group
Case-series	Descriptive	Retrospective	Observational	Disease	No
Cross-sectional	Descriptive	Retrospective	Observational	Both simultaneously	Usually not
Case-Control	Analytic	Retrospective	Observational	Disease	Yes
Cohort	Analytic	Prospective and Retrospective	Observational	Usually exposure/cause	Usually Yes (may be integral to the study population)
Experimental/ Intervention	Analytic	Prospective	Experimental	Usually disease, but sometimes cause of disease	Yes, with exceptions

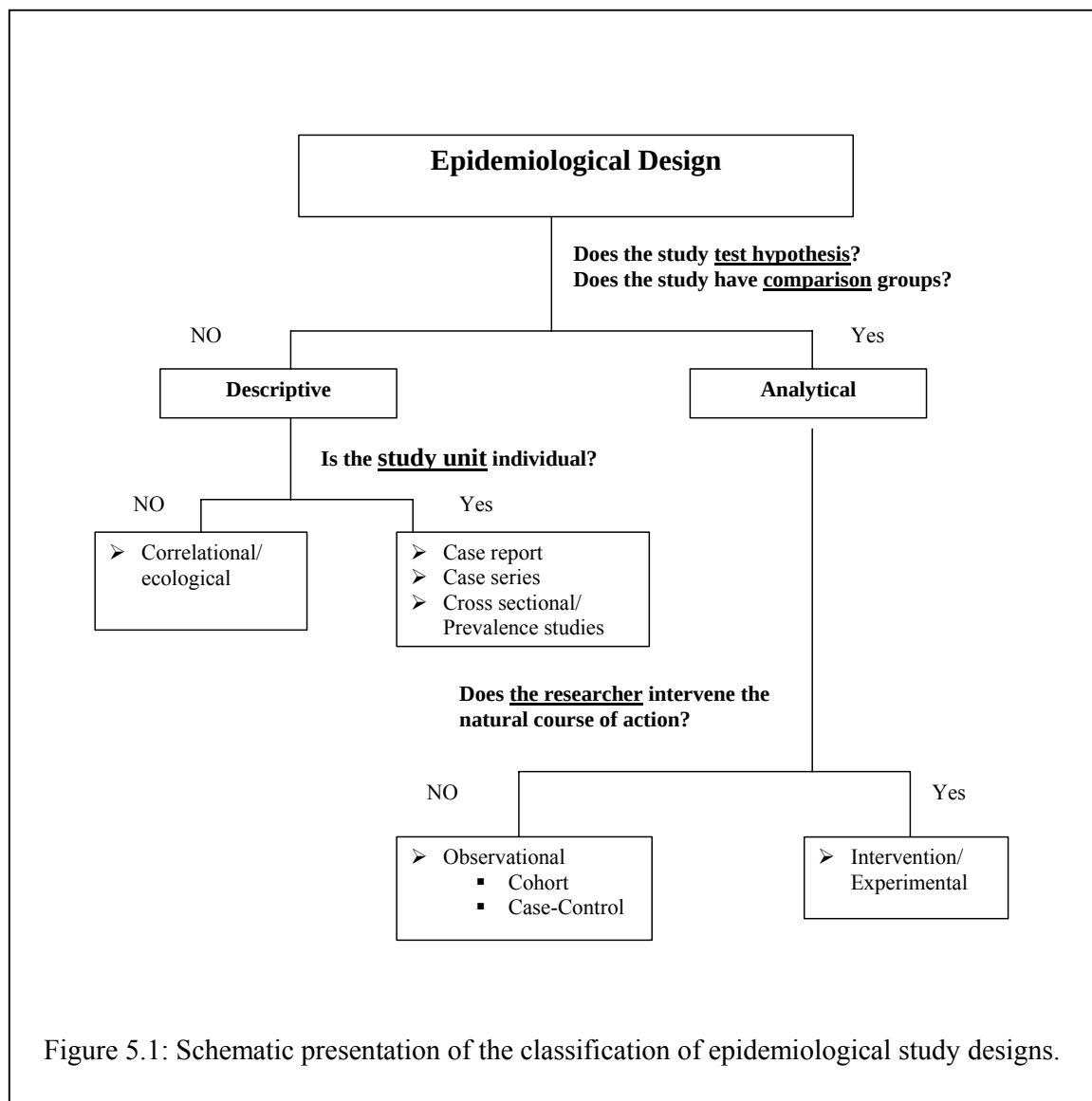


Figure 5.1: Schematic presentation of the classification of epidemiological study designs.

5.A. Descriptive studies

Some of the important features described below shows the commonness and usefulness of descriptive studies in improving health services and promoting health research. Descriptive studies:

- are mainly concerned with the distribution of diseases with respect to time, place and person.
- provide useful information for health managers to allocate resource and to plan effective prevention programmes.
- generate epidemiological hypothesis, an important first step in the search for disease determinants or risk factors.
- can use information collected routinely which are readily available in many places. So generally descriptive studies are less expensive and less time-consuming than analytic studies.
- are the most common type of epidemiological design strategies in medical literature.

There are three main types of descriptive studies, which are discussed in detail below:

- Correlational/ecological
- Case report or case series
- Cross-sectional

5.A.1. Correlational or Ecological

- Uses data from entire population to compare disease frequencies - between different groups during the same period of time, or in the same population at different points in time.
- Does not provide individual data, rather presents average exposure level in the community.
- Cause could not be ascertained.
- Correlation coefficient (r) is the measure of association in Correlational studies. It is important to note that positive association does not necessarily imply a valid statistical association.

e.g. Hypertension rates and average per capita salt consumption compared between two communities.

Average per capita fat consumption and breast cancer rates compared between two communities.

Comparing incidence of dental cares in relation to fluoride content of the water among towns in the rift valley.

Mortality from CHD in relation to per capita cigarette sales among the regions of Ethiopia.

Strength: can be done quickly and inexpensively, often using available data.

Limitation:

- i. Inability to link exposure with disease. Data on exposure and outcome are not linked at the individual level; association found with aggregate data may not apply to individuals (this is referred as ecological fallacy). For example, in the association between high fat intake and breast cancer it is difficult to know whether the risk is higher among individual women who have high intake of fat. In the association between reduced mortality from cervical cancer and PAP smear screening, it is difficult to know whether the reduction is really in those women who were screened by PAP smear or otherwise.
- ii. Lack of ability to control for effects of potential confounding factors. There may be other things that are the true cause. For example, often people with high fat consumption also have high meat consumption. Perhaps it is the meat that is actually responsible for the breast cancer - or may be because of reduced vegetable intake or merely a reflection of socio-economic status. Another example is the correlation found between the high per capita colour TV and mortality from CHD, again here it is obvious that colour TV owning is not a good reason for increased mortality from CHD.
- iii. It may mask a non-linear relationship between exposure and disease. For example alcohol consumption and mortality from CHD have a non-linear relationship (the curve is "J" shaped), but this type of relationship is impossible to demonstrate in Correlational studies.

5.A.2. Case Report and Case Series

Describes the experience of a single or a group of patients with similar diagnosis or health problem derived from either the practice of one or more health care professionals or a defined health care setting such as hospital, health centre or specialised clinic. Often it has limited value due to the limitations mentioned below, but occasionally could be revolutionary. The following information/data are important inputs in making better use of the case series study:

1. Defining the disease or health problem clearly
2. Recording the date when the disease/death occurred (Time)
3. Recording where the person lived, worked,... (Place, information relevant to the study with regard to place)
4. Recording personal characteristics of the person such as age and sex (Person)

5. Explore the opportunities for collecting additional data from records or the person directly.
6. Estimating the size and the characteristics of the population at risk.

E.g. The 5 young homosexual men with PCP seen between Oct. 1980 and May 1981 in Los Angeles created a serious concern among physicians since PCP among young adults is not common. Later, with further follow-up and thorough investigation of the strange occurrence of the cluster of cases the diagnosis of AIDS was established for the first time.

One case of pulmonary embolism observed 5 weeks after a woman started using oral contraceptive was the first clue to the association between oral contraception and increased risk of venous thromboembolism, an established fact today.

Strength:

- useful for studying signs and symptoms and creating case definitions for epidemiological studies
- case-series that include cases at various stages of an illness from mild cases to dead supplemented by investigation of the past medical history of these cases and observing them to death (doing autopsy as appropriate) can help build up a picture of the natural history of a disease.
- very useful in providing critical information, for hypothesis generation, for sound analytical studies.

Limitations:

- Report is based on single or few patients, which could happen just by coincidence.
- Lack of an appropriate comparison group.
- Rates can not be calculated since the population corresponding to the source of cases can not be defined well.
- Detailed and complete risk factor information is difficult to obtain for all cases from records.
- Studies are prone to atomistic fallacy (the opposite of ecological fallacy); the forces that cause or prevent disease at an individual level are different from those that work at societal level. For example, at an individual level a high income may be associated with lower rate of suicide but this does not mean that societies which are rich have a lower rate of suicide or better mental health.

5.A.3. Cross Sectional Studies (Survey)

Cross sectional study investigate disease and risk factor (exposure) patterns in a representative sample of a population in a narrowly

defined time period. An ideal cross-sectional study is done on a geographically defined population. It can be useful to identify associations, generate and test hypothesis and, by repeating at different time periods, measure change and hence evaluate interventions. Comparison between subgroups within the sample or deliberately designed comparison groups can be made. Findings from cross-sectional study of a sample population can be generalized cautiously if basic characteristics of the populations are similar. Cross sectional studies may be snapshot (done in a day) but often made over a period of time that extends from a few days to several years. Studies done over a period of a year escape the problem of seasonal variations since it covers all seasons. Studies involving rapidly changing phenomena need to be conducted quickly and repeated to give useful results. Cross sectional studies provide the most reliable estimate of burden of disease in a population by estimating prevalence. For factors that remain unaltered overtime, such as sex, race or blood group, the cross-sectional survey can provide evidence of a valid statistical association. However, cross sectional studies are useful in raising the question of the presence of an association rather than testing hypothesis.

Strength:

- Easy to conduct
- Not time consuming
- Can be used to compare population with different characteristics as in comparative cross sectional studies

Limitation:

- "chicken or egg" dilemma - difficult to know which occurred first, the determinant/exposure or the outcome. Therefore, difficult to distinguish whether the exposure preceded the development of the disease or whether presence of the disease affected the individual's level of exposure

E.g. In the study of knowledge of modern contraceptive, and use of contraception, you may show that women who know about modern contraception are more likely to use it. So you may want to educate women about it, believing that this will lead to higher rate of use. The problem is, did the women know about it and then start to use it, or did they learn about it because they were using it?

Another example is community health agent activity and health station supervision. Are the CHAs active because they are supervised, or do the health stations supervise CHAs that they know are doing something. There are factors which clearly come before the outcome of interest in time. For example, if more women CHAs are active than men CHAs, one can be sure

that their sex came before their activity as a CHA in time, and thus it is their sex that causes them to be active, not their activity which cause them to be female.

- Survivor bias- people who died of the disease are missed in cross-sectional study. One way of correcting this problem is to supplement population studies with clinical studies.

5. B. ANALYTIC STUDIES

Focuses on the determinants of a disease by testing the hypothesis formulated from descriptive studies, with the ultimate goal of judging whether a particular exposure causes or prevents disease. Analytic studies are broadly classified into two - observational and interventional studies. Both types use "control group", the use of control group is the main distinguishing feature of analytic studies. Following a brief presentation of the classification of analytic studies each will be discussed detail.

5.B.1. Observational studies

Information are obtained by observation of events. No intervention is done, no deliberate interference with natural course of disease. Cohort and case-control are in this category.

i. Cohort

Subjects are selected by exposure, or determinants of interest, and followed to see if they develop the disease or outcome of interest.

E.g. Take Awrajas with trained manager and untrained managers and follow them to see which group will do better to increase coverage.

Follow 100 children who received BCG vaccination and another 100 who didn't get BCG vaccination and see how many of them get tuberculosis.

ii. Case Control

Subjects are selected with respect to presence or absence of disease, or outcome of interest, and then inquiries are made about past exposure to the factor(s) of interest.

E.g. Take people with and without TB, ask them if they ever had BCG vaccination.

Take Awrajas with high and low EPI rates, ask them if their Awraja health managers were trained.

5.B.2. Interventional / Experimental

- The researcher does something about the disease or exposure and observes the changes.

- Investigator has control over who gets exposure and who don't. The key is that the investigator assign into either group, whether it is done randomly or not.
- Always prospective.

E.g. Assign children randomly to get chloroquine or not, and see how many develop symptomatic malaria.

5.C. CASE-CONTROL STUDIES

Epidemiologic research methods in which the two study groups are selected on their disease status. This is a design strategy developed in response to the difficulty of studying diseases with very long latency period. The design is capable of evaluating the association of a disease to exposure many years after the actual exposure. Because of this and its efficiency in time and cost case-control studies have become the most common analytic design encountered in medical literature. The prototype study on lung cancer and smoking was done in 1950's. The word case is related to the outcome of interest in the study, which commonly comprises individuals with the health problem of interest. The comparison group (control, referent) supplies information about the expected risk factor pattern in the population from which the case group is drawn. Of the epidemiological designs, case control is the most focused on establishing causation and least on measuring burden of disease or risk factors.

Design and conduct of case-control studies

In the design of the study always seek for the comparability between cases and controls; this is the basis for valid conclusion.

Defining Cases:

Establish a clear operational definition or use standard definition of disease (outcome) of interest in order to have a clear understanding of exposure-disease association.

E.g. "Uterine Cancer", before 1940 include Ca of the body of uterus and Ca of the cervix.

"congenital malformation" and drug use - specify malformation

==> John's criteria to diagnose rheumatic heart disease is a good example.

If you are not certain about the diagnosis, and if the information collected is adequate perform analysis separately for cases classified as definite, probable or possible.

Selection of Cases:

Do not always go for random representation of cases, rather it is better to restrict yourself to cases on which you can get complete and reliable information. In order to evaluate the public health importance of the findings it is important to know whether the cases are typical of all cases of the disease of interest. Select controls which are comparable to the cases entered into the study, do not try to

represent the population of all non-diseased persons. It is important to know the geographic area and the time period when the cases occurred to draw an appropriate control group for the study; difference in healthcare practice and policy, and population behaviour could differ in geographic areas and by time period.

Hospital-based Vs population-based cases

Hospital-based: easy and inexpensive to conduct but it is prone for selection bias.

Population-based: avoids selection bias, allows the description of a disease in the entire population and the direct computation of rates of disease in exposed and non-exposed persons.

Incident Vs Prevalent cases

The ideal set of cases would be new (incident) and representative of all cases of the health problem under study.

Prevalent cases:

- Increase sample size available for rare disease.
- Difficult to establish temporal sequence between exposure and outcome. E.g. Coffee consumption and peptic ulcer disease.
- Use is unavoidable in certain situations, like in studying congenital malformations which are rare to find.

Incident cases:

- Helpful to establish temporal relationship between exposure and outcome. So, it is better to limit cases to those newly diagnosed within a specified period of time if the aim is also to establish temporal relationship.
- Records are easily obtainable and recall is not a serious problem.

Selection of controls:

There is no control group that is optimal for all situations. Controls are made for a particular group of cases, do not try to represent the entire non-diseased population rather try to achieve comparability between the cases and controls. Selection of controls should consider besides comparability, practicability and economic impact.

The control series is intended to provide an estimate of the exposure

rate that would be expected to occur in the cases if there were no association between the study disease and exposure.

Sources of controls:

1. Hospital Controls

Advantages:

- Easily identified and readily available in sufficient number with reduced cost than population controls.
- More likely than healthy individuals to be aware of antecedent exposures or events --> minimize recall bias.
- Controls are also likely to have been subject to the same intangible selection factors that influence cases to come to this particular physician or hospital --> minimize selection bias
- More likely to be cooperative because they anticipate benefit from their involvement or might think that its related with their illness --->reduce bias due to non-response.

Disadvantages:

- Because they are ill they are different from healthy individuals in many ways. Several studies in the West have demonstrated that hospitalized patients are more likely to smoke cigarette, use oral contraceptive, and be heavy drinkers of alcohol than non-hospitalized individuals.
- There is danger of altering the direction of association or masking a true association between exposure and outcome of interest. Patients with diseases known to be associated either positively or negatively, with the exposure of interest, should be excluded from the control series. For example, in studying the association of cigarette smoking and lung Cancer, individuals with other respiratory illnesses could not be taken as controls, since smoking is also known to have some association with other respiratory illnesses.

2. General population controls

Advantages:

- Generalizability is possible
- Good when cases are selected to represent affected individuals in a defined population. For example, if cases to that particular hospital are coming from a geographically defined area selection of controls from the entire population could be possible.

Disadvantages:

- Costly and time-consuming
- Recall bias - controls may not recall exposures with the same level of accuracy since they may not be seriously concerned about their illness.
- People might be less motivated to participate for the same reason given above, which increases non-response rate, i.e., selection bias.

3. Special controls

Special controls are individuals which are related to the cases in some way. These are friends, household members (siblings,...), neighbours,...

Advantages:

- they are healthy.
- more likely to be cooperative than members of the general population, because of their interest in the cases.
- offer a degree of control over some confounding factors, such as ethnicity, socioeconomic status, or environment.

Disadvantage:

- if the study factor is likely to be similar to the cases, an underestimate of the true effect of the exposure of interest may result. E.g. if the study factor is diet, it will be similar for both cases and controls, if controls are siblings.

Number of control groups and case-control ratio

A single control group is optimal in most of the times. Add more control groups only when you are not confident with the control group or when you see a clear deficiency in your control group or when there is a clear advantage by adding another control.

Conditions for multiple controls:

- when the control is not considered appropriate.
- when the selected group has a specific deficiency that could be overcome by inclusion of another control group.

Control-case ratio

The optimal control-case ratio is 4:1. As the number of controls per case increases, the power of the study also increases. But, beyond 4:1, there is only a small increase in statistical power, which can not justify the expenditure of additional resources.

Ascertainment of disease and exposure status

Potential source of information must be carefully considered in terms of its ability to provide accurate as well as comparable information for all study groups. Procedures used to obtain information must be similar for cases and controls:

- place and circumstances of interview must be the same.
- blind interviewers or record reviewers, if possible.
- data collectors should be unaware of the specific hypotheses being tested --> to reduce observation bias.
- the ability to obtain exposure information from records completed before the occurrence of outcome events is especially valuable. E.g. record of X-ray during pregnancy in studying its effect on the child (congenital malformation).
- ascertainment of exposure should involve defining the part of a person's exposure history that could be relevant to the aetiology of the disease under study.

E.g. Smoking & lung Ca - duration of smoking is important than the amount currently smoked.

Smoking & Myocardial Infarction - current smoking is most important.

==> Collect information in such a way that it allow you to identify the most appropriate time window for the evaluation of the possible harmful effects of an exposure - try to avoid collecting information over too wide a period, such as "ever use" in order to avoid the inclusion of some period in time that cannot be causally related to the disease.

Issues in analysis

Comparison is made primarily by estimating the relative risk as computed by the odds ratio. If Case Control study is population based, or if estimates of disease incidence are available from an outside source, rates of disease for the exposed and non-exposed can be computed and compared directly.

Odds ratio can provide a valid estimate of the relative risk if the following assumptions are fulfilled:

- the cases are incident cases drawn from a known and defined population;
- the controls are drawn from the same defined population and would have been in the case group if they had the disease;
- controls are selected in an unbiased way, e.g. independently of exposure status; and
- if the disease is rare.

5.D. COHORT STUDIES

Epidemiologic design in which the two comparison groups are defined according to their exposure status to a suspected risk factor of a disease is cohort. The two groups should be free of the study outcome. The main feature of a cohort study is observation of sufficiently large number of persons over a sufficiently long period of time to generate reliable incidence or mortality rates in the population subsets. Unlike cross sectional study the health outcome of health change data are obtained on the same individuals in a population at more than one time. Cohort study starts by establishing baseline data, usually from cross sectional survey, or less commonly by extracting relevant information from census or routine information systems. One of the main functions of cohort study is to provide information on the incidence and to describe natural history of disease. If the cohort study is based on a defined and characterized population the incidence rates can often be generalized to similar populations.

It is important to properly differentiate cohort and cohort study as defined below:

Cohort: is a group of persons with common characteristics, usually an exposure or involvement in a defined population group, who are followed or traced over a period of time..

Cohort study (synonyms: concurrent, follow-up, incidence, longitudinal, prospective study): is the analytical method of epidemiologic study in which subsets of a defined population can be identified who are, have been, or in the future may be exposed or not exposed, or exposed in different degrees, to a factor or factors hypothesized to influence the probability of occurrence of a given disease or other outcome.

Types: There are two types of cohort studies, prospective and retrospective, depending on the temporal relationship between the initiation of the study and the occurrence of the disease. The design concepts of the two types of studies is illustrated in Figure 5.2.

1. Prospective - At the beginning of the study the outcome has not yet occurred. Regarded as more reliable than the retrospective, if the sample size is large and follow-up complete.

* *The outcome has not occurred at the beginning of the study*

2. Retrospective - Both exposure and outcome status have occurred at the beginning of the study. This is possible where medical records permit accurate assessment of both risk factors and disease outcomes, in which case a retrospective cohort study is possible without any prospective work. Efficient in cost and time. Often uses of data collected for other purposes, so

information obtained might be incomplete and non-comparable for all subjects.

** Both exposure and outcome have occurred before the beginning of the study*

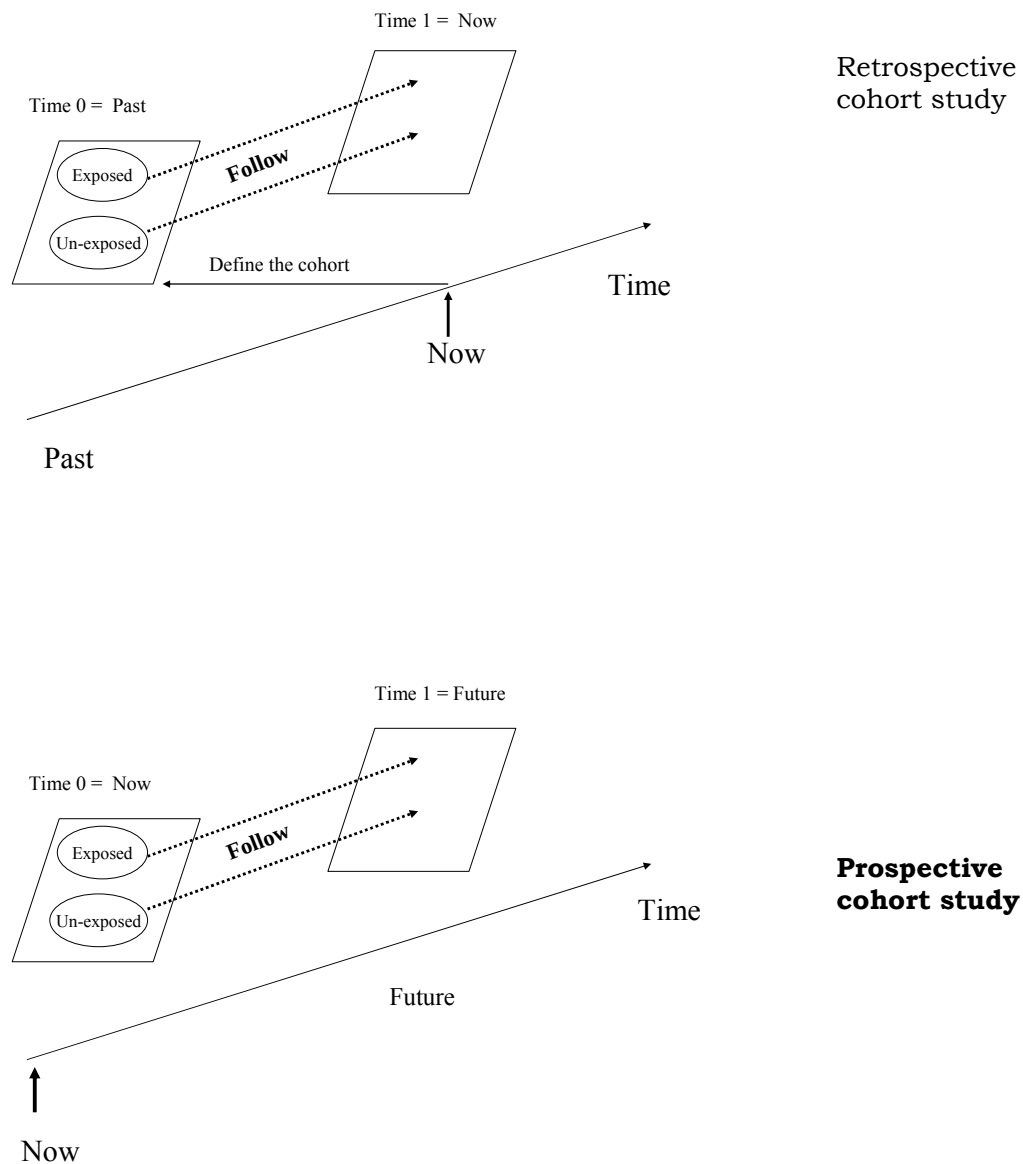


Figure 5.2. Population Design concepts of Cohort Studies

Selection of Exposed Group

selection of exposed group should consider scientific and feasibility issues which include:

- the frequency of the exposure of interest in the study population.
- the need to obtain complete and accurate exposure and outcome information on all study subjects. Example: the use of physicians or nurses permits longer and fairly complete follow up.
- the ability of obtaining sufficient exposed individuals in a reasonable period of time - identify high risk population (special group) to the exposure of interest.

Selection of high risk group also allows the evaluation of a rare disease. Although cohort studies are in general not optimal for the evaluation of rare diseases, if the outcome of interest is relatively common among those exposed; i.e., if the attributable risk percent is high the design can be used efficiently.

- the ease to collect relevant information and to follow-up.

Selection of controls

Always attempt to select a control group which is comparable to the characteristics of the exposed population. There is no single optimal control group that can be used for any circumstance.

Source of data

The major consideration should be the availability of accurate and complete information on exposure and outcome of interest in the study groups in a way that is comparable to both.

Exposure ascertainment:

1. Using **Pre-existing records**: from hospital, employers record..

Advantages:

- can make available information for high proportion of cohort.
- relatively inexpensive to obtain.
- allow objective and unbiased classification of exposure status.

Disadvantages:

- information on exposure level may be insufficient.
- may not contain adequate information on potential confounders.

2. By conducting **Interview** and filling **questionnaire**

Advantages:

- enables to record exposure information that are not routinely recorded, particularly lifestyle factors.

Disadvantages:

- potential for information bias, particularly recall. In such situations, where objective sources can not be used, it is important that information is obtained in a comparable manner for all participants.

Outcome ascertainment:

With adequate consideration to the resources available for the study, the *aim is to obtain complete, comparable and unbiased information on the subsequent health experience of every study subject*. One or a combination of the following sources could be used: routine surveillance, death certificate, periodic health examination, autopsy records, hospital records, etc.

Always try to have a *firm outcome criteria and standard diagnostic procedure* which are equally applied for exposed and non-exposed individuals. Do not do any diagnostic examination only for one group, because the difference which might be observed could be just due to the greater opportunity offered to be diagnosed.

Follow-up

This the *major challenge* in cohort studies, as well as the major cost in terms of time. Unless complete or nearly complete information could be obtained the results might be un-interpretable. If the loss to follow-up is not comparable between the two exposed groups, this will also be a source for bias. Therefore, if there is a need for long follow-up period, the mechanism to achieve complete follow-up should be thought carefully in the planning of the study.

Analysis

The basic analysis in cohort studies are:

- calculation and comparison of rates of the incidence of the outcome for exposed and non-exposed.
- comparison of the two groups with baseline characteristic to ensure similarity.

Issues in interpretation of cohort studies

Role of bias :

Misclassification bias to some extent might be unavoidable. So, always attempt must be done to avoid the introduction of any systematic misclassification.

Random misclassification or error unrelated to the outcomes of interest may not affect comparability, rather it dilute or underestimate any true association that may exist between the exposure and outcome. As a result, the observed RR estimate will always be biased towards the null value of 1. On the other hand differential misclassification can result in a biased risk estimate that is either an underestimate, an over estimate, or, by chance, the same as the true measure of association.

Effects of losses to follow-up:

If the probability of loss is related to exposure, outcome or to both, or if the proportion is large the estimate of exposure-disease association may be biased.

Because of the difficulty to know which factors are related to loss, the best way to eliminate bias is by reducing loss to follow-up to an absolute minimum.

For losses:

- try to get at least mortality status from other sources.
- examine previously collected data to determine whether there are systematic differences between the losses and follow-ups.
- indirectly calculate exposure-disease association, assuming the two extreme outcomes. One assuming all those who were lost to follow-up developed the outcome of interest and the other assuming that none developed the outcome - this provides a range within which the true association will lie. If losses to follow-up are large, the observed range will be so wide as to provide little useful information.

Effect of non-participation

This does not affect validity unless non-response is related to both the exposure and other risk factors for the outcome under study. The effect of the difference is mainly on generalizability of the study results.

The possible effect of non-response on either generalizability or validity can be assessed by comparing basic social and demographic characteristics of those who do and do not participate in a study.

Table 5.3 Advantages and Limitations of Cohort and Case-Control Study Designs.

Case-Control	Cohort
<u>Advantages</u> • optimal for the evaluation of RARE diseases • can examine multiple etiologic factors for a single disease • quick and inexpensive • relatively simple to carry out • guarantee the number of persons with cases	• valuable when the exposure is rare • can examine multiple effects of a single exposure • can elucidate temporal relationship • allows direct measurement of risk • minimize bias in ascertainment of exposure
<u>Limitations</u> □ inefficient for the evaluation of rare exposure □ can not directly compute risk □ difficult to establish temporal relationship □ determining exposure will often rely on memory □ persons who die as a result of disease caused by the determinant may not be known to the study	□ inefficient in evaluation of rare diseases □ expensive □ time consuming □ loss to follow-up create problem

Summary of Measures of Association: case-control studies

Odds of disease : is a simple ratio, not a proportion.

Odds of disease in exposed = a/b

Odds of disease in unexposed = c/d

Relative Odds or Odds Ratio (OR)

$$OR = \frac{ad}{bc}$$

Attributable Risk Percent (AR%) among exposed:

$$AR\% = \frac{\text{Risk in the exposed} - \text{Risk in unexposed}}{\text{Risk in exposed}}$$

$$= \frac{OR - 1}{OR} \times 100$$

Population Attributable Risk Percent (PAR%) :

$$PAR\% = \frac{\text{Risk in population} - \text{Risk in unexposed}}{\text{Risk in population}}$$

***Attributable risk (AR) or Risk difference (RD)**

$$AR = \text{Risk in exposed} - \text{Risk in non-exposed} \\ = a/a+c - c/c+d$$

***Population attributable risk (PAR)**

$$PAR = R \text{ total population} - \text{Risk in unexposed} \\ = AR \times \text{proportion of exposed individuals in population}$$

*** If the study is population based or if incidence rates can be estimated.**

Summary of Measures of association: cohort studiesAttributable risk (AR) or Risk difference (RD)

$$AR = \text{Risk in exposed} - \text{Risk in non-exposed}$$

$$= a/a+b - c/ c+d$$

Relative risk (RR) or Risk Ratio

$$RR = \frac{\text{Risk in exposed}}{\text{Risk in unexposed}}$$

Attributable Risk Percent (AR%) among exposed:

$$AR\% = \frac{\text{Risk in the exposed} - \text{Risk in unexposed}}{\text{Risk in exposed}}$$

$$= \frac{AR - 1}{I_e} \times 100$$

Population attributable risk (PAR)

$$\begin{aligned} PAR &= \text{Risk total population} - \text{Risk in unexposed} \\ &= AR \times \text{proportion of exposed individuals in population.} \end{aligned}$$

Population Attributable Risk Percent (PAR%) :

$$PAR\% = \frac{\text{Risk in population} - \text{Risk in unexposed}}{\text{Risk in population}}$$

$$= \frac{PAR}{\text{Incidence rate of disease in population}} \times 100$$

5.E. INTERVENTION STUDIES

This is an epidemiological design that closely resembles the controlled experiment in basic science researches, and can produce high quality data if done properly. The main distinction from other types of analytic studies is that individuals are allocated into experiment or control group by the investigators. It has essentially the same design as prospective cohort study with one very key difference, exposure status of the study population is deliberately changed by the investigator to observe how this alters the incidence of disease or other features of the natural history. Intervention trials could be done for various purposes:

- Proof of concept trial: designed solely to produce knowledge about cause and effect, does not test the efficacy of the intervention in actual practice.
- Prevention trial: interventions are to prevent disease and study participants are persons without disease.
- Clinical trial: interventions are treatment based on drugs and study participants are persons with disease.

The comparison groups in intervention study are known as the intervention group and the control group. The intervention group receives the test drug (the preventive activity such as health education, diet and exercise). The control group shall be offered the best known alternative or a placebo activity with no known effect on the outcome. It is very important that the two groups gain equal amount of attention in the study. Unequal attention leads to differences attributed to the amount of attention each group receives, not to the intervention- known as a Hawthorne effect (bias). Ideally, the intervention and the control populations are at the same stage of the natural history of the disease and are similar in the characteristics that affect disease outcomes, differing only in the exposure of interest to the intervention study.

Classification

1. Based on population

- A. clinical trial - usually performed in clinical setting and the subjects are patients.
- B. Field trial - used in testing medicine for preventive purpose and the subjects are healthy people. E.g. vaccine trial
- C. Community trial- unit of the study is group of people/community. E.g. fluoridation of water to prevent dental caries.

2. Based on design

- A. Uncontrolled trial - no control group. control will be past experience (history).
- B. Non-randomized controlled- there is control group but allocation into either group is not randomized.

- C. Randomized controlled - there is control group and allocation into either group is randomized.

3. Based on objective

- A. Phase I - trial on small subjects to test a new drug with small dosage to determine the toxic effect.
B. Phase II - trial on small group to determine the therapeutic effect.
C. Phase III- study on large population - usually a randomized control trial.

Problems Related to Intervention Studies

1. Ethical considerations prevent evaluation of many treatments or procedures using an intervention design strategy.

Some of the ethical issues are:

- Practices or substances already known to be harmful should not be used in this study.
- Therapies known to be beneficial should not be withheld from any affected individuals in the study population.
- Investigators have to have a complete knowledge of the subject under study.
- The researcher must have at least informed consent from each study participant and subjects should be left free to withdraw from the study at anytime.
- A written research protocol is a must.

2. Feasibility/ practical issues

- Subject recruitment, getting adequate individuals to enrol into a study is not easy.
- Conducting trial on a widespread practice poses difficulty in getting sufficiently large population who are willing to undergo through a new treatment or practice believed to be more beneficial than the old treatment or practice for the duration of the entire study period; i.e., its difficult to achieve satisfactory compliance from all study subjects for a long time, particularly if study period is quite long. Getting an appropriate control group is also difficult sometimes. For example, if you want to see the association of chat chewing and dental caries and the prevalence of chat chewing is 70%, it will be difficult for you get an adequate control.

3. Cost - experimental studies are often very expensive because of the long follow-up period, which is comparatively longer for preventive trials, and arrangements for follow up outside the clinic settings.

Issues in the design and conduct of clinical trials

Intervention studies to represent the "gold standard" for epidemiologic research should consider the following:

1. Selection of a study population

Reference population: The general group to whom investigators expect the results of the particular trial to be applicable. Represents the scope of the public health impact of the intervention. And, it is related to the issue of generalizability.

Experimental population: The actual group in which the trial is conducted. It is preferable that if this group is not different from the reference population for the sake of generalizability, but this should not be a concern.

Considerations in choosing the experimental group:

- ☐ Check whether the proposed experimental population is sufficiently large to achieve the required sample size for the trial.
- ☐ Choose population that will experience a sufficient number of endpoints to permit meaningful comparison.
- ☐ likelihood of obtaining complete and accurate follow-up information for the period of trial.

2. Allocation of study groups

Allocation into either group must be done after determining eligibility and getting consent. It is always advantageous to do the allocation at random.

Randomization: can be done using random-number table or using small computers which are capable of generating random numbers. If the sampling frame is small a lottery method can be applied.

Advantages:

- . Treatment groups will not be known by the researcher.
- . "On average" the study group will be comparable; i.e., known and unknown potential confounders will be equally distributed between the two groups.
- . Randomization can provide a degree of assurance about the comparability of the study groups that is simply not possible in any observational design.
- . The impression it poses on the readers (consumers) - less proof is

needed to show that the observed result is due to a selection bias or confounder effects.

Maintenance and assessment of compliance

Subjects may decline from the treatment protocol for various reasons after randomization, and this related to the length of time that subjects are expected to adhere to the intervention, as well as to the complexity of the study protocol. It is always important to obtain as complete follow-up information as possible since they will be included in the primary analysis.

Methods to enhance compliance:

- . select population who are both interested and reliable.
- . arrange frequent contacts with individuals
- . use incentives, such as providing medical information

Assessment of compliance:

Non-compliance will decrease the statistical power of a trial to detect any true effect of the study intervention. Therefore, to see its effect compliance levels in any study must be measured. Measuring compliance is not easy, and all the measures available have inherent limitations. Some of the measures are:

- . Self-report, the simplest and the only way to assess behavioural modification and exercise programs.
- . Pills count - ask participants to bring unused pills to each clinic visit, this may eliminate inaccuracies due to poor memory, it assumes that all the unreturned pills has been ingested.
- . Biochemical tests
 - . used to validate self-report
 - . objective but expensive and logistically difficult - Riboflavin is a safe biochemical marker that has been used added in the treatment. But, can only reflect the ingestion of the pills the preceding day or two and thus can not be used as reliable measure for long-term compliance.

It is inevitable that some portion of participants in a trial will become noncompliant despite all reasonable efforts. In any case, it is important to obtain as complete follow-up information as possible since they will be included in the primary analysis.

Ascertainment of outcome

Use uniform ascertainment of outcome for complete follow-up period for all study subjects. To eliminate a possible bias, maintain a high level of follow-up and reduce the proportion of outcomes that are not ascertained to the minimum and comparable between the two groups. Follow-up is short in assessing the effect of acute disease and long in assessment of chronic disease outcomes. The difficulty in maintaining complete ascertainment of outcome increases with increasing length of follow-up.

Potential for observation bias in ascertainment of outcome can exist in an intervention study in that knowledge of a participant's treatment status might, consciously or not, influence the identification or reporting of relevant events. This can be overcome by the use of placebo and blinding.

Placebo - an inert agent indistinguishable from the active treatment. Use of placebo minimizes bias in the ascertainment of both subjective disease outcomes and side effects.

Placebo effect: tendency for individuals to report favourable response to any therapy regardless of the physiologic efficacy of what they received.

The use of placebo ensures that all aspects of the intervention offered to participants are identical except for the actual experimental treatment. With no placebo, it is impossible to tell whether subjective outcomes are due to the actual trial treatments, to the extra attention participants receive, or merely to their belief that the treatment will help.

The primary strength of a double-blind design (study subjects and health care giver do not know who is getting the active intervention) is to eliminate the potential for observation bias. Of course, a concomitant limitation is that such trials are usually more complex and difficult to conduct. Circumstances in which double-blinding is not possible are evaluation of programs involving substantial changes in life-style, such as exercise, cigarette smoking or diet, surgical procedures, or drugs with characteristic side effects.

A triple-blind trial is where the study subject, the field investigator and the health care provider do not know who is receiving the active treatment. This is even more complex than the double blind study and requires complicated procedures to safeguard the safety of study subjects.

Problems associated with unblinded trials:

- subjects who are not on the new or experimental program may become dissatisfied and dropout of the trial, thus resulting in differential compliance or loss to follow-up.
- knowledge of the intervention to which group the participant has been assigned might raise the potential for observation bias in the reporting of side effects or assessment of outcome.

The quality of "gold standard" in intervention studies can be achieved through :

Randomization
Use of placebo
Double Blinding

Stopping Rules: Decision for early termination of a trial

To assure the welfare of the participants is protected, interim results should be monitored by a group that is independent of the investigators conducting the trial. Consider termination if the interim results indicate a clear and extreme benefit on the primary end point due to intervention, or if one treatment is clearly harmful. It would also be unethical to stop a trial prematurely based solely on emerging trends from a small number of patients - the aim must be to achieve an equitable balance between, on the one hand, protection of randomized participants against real harm and, on the other, minimizing the risk of mistakenly modifying or stopping the trial prematurely.

Requirements for modification or termination of an ongoing trial:

- 1st- Observation of a sustained statistical association that is so extreme, and, therefore, so highly significant, that it is virtually impossible to arise by chance alone.
- 2nd- consider the observed association in the context of totality of evidence:
 - . Is there known or postulated biologic mechanisms that might explain the observed effect?
 - . Is it in-line with other randomized trials, or those from observational studies?
 - . How does the observed association(effect) affect the risk-to-benefit ratio of the intervention?

POWER OF THE STUDY

The statistical power of a trial to detect a postulated difference between treatment groups, if one truly exists, is dependent on:

1. Sample Size

Trials with inadequate sample size might have a great potential for scientific harm - could be as a result of misinterpretation. Always its advisable to take sample large enough to detect small to moderate (10-20%) benefit or differences that resulted from the intervention.

2. Accumulation of adequate end points

There are at least two major strategies to obtain adequate numbers of end points:

a. Selection of a high-risk population

The collection of baseline data can be planned to allow the identification of particular subgroups who might experience the effects of an intervention more than others; i.e., those at a higher risk of developing the outcome of intervention.

b. Length of follow-up period

It is always better to consider that the actual rate of occurrence of end points will be less than the projected level, which could be due to the low incidence of the outcome of interest in the volunteer study population, this is referred as "healthy volunteer effect"- the only way to compensate for this deficit is to extend the length of follow-up to get more events.

Secular changes in disease rates during the course of the trial might be sometimes as great as that due to the intervention. E.g. During the decade in which MRFIT trial was conducted, the entire U.S population including all MRFIT participants, experienced a marked 25 to 30% decline in Coronary Heart Disease(CHD) mortality. As a result, the expected numbers of deaths in the trial was less by two-third, so the follow-up was extended to increase the number of end-points (the outcome).

Consider the postulated mechanism by which the study agent (the exposure) exerts its effect in deciding the length of follow-up period. That is, how long will it take for the study agent to exert its effect on the end result.

Every effort should be made to incorporate an adequate length of follow-up during the planning phase of the trial. IF, for any

reason, there is a need to alter the follow-up period, the decision should be made as early in the trail as possible to maintain the scientific credibility of the study and avoid the implication that the change in study design was based on last-minute efforts to achieve statistical significance.

3. Effect of Compliance

Compliance must be assessed in all study participants, regardless of their particular treatment assignment. The effect of non-compliance in any participant is to make the intervention and comparison groups more alike, as a result decreases the ability of the trial to detect any true differences between the groups.

One strategy to increase compliance is to use "Run-in or Wash out" period prior to the actual randomization- all participants receive either the active treatment or the placebo for a number of weeks or months before formal randomization to a treatment group. The only limitation to this strategy is the limitation to generalize study into reference or general population, but the primary goal of a trial must be to attain a valid result.

Issues in Analysis and Interpretation of Intervention Studies

Basically the issues of analysis in intervention studies are the same as that of the analysis of cohort studies. The fundamental comparison to estimate the true benefit of the intervention program should be obtained through analysing the data by intention to treat - "once randomized, always analyzed"- so always maintain high level of compliance, keep losses to follow-up at a minimum, and collect information on all randomized subjects.

Reasons:

1. Non-compliance may be related to factors that also affect the risk of the outcome under the study, and failure to analyze data on all randomized participants could introduce bias. In most studies, perfect compliers represent only a fraction of the total study population.
2. Analysis of compliers data does not address the actual research question posed in an intervention study. First, it is only the entire groups allocated by randomization that are truly comparable - so preserve the power of randomization by analysing the entire population. Secondly, if a particular regimen is so difficult and uncomfortable that it is likely to be accepted and used by only a small proportion of the reference population, it may not be practical to recommend its use, no matter how effective the actual treatment may be.

However, subsequent analysis can be performed based on that subgroup of participants who actually received their assigned treatment. However, even if it is possible to perform analysis that achieve balance in the distribution of known confounders, it is impossible to regain the control of unknown confounders that had been achieved originally through randomization.

3. Rule out other possible alternative explanations for the observed findings. Alternative explanations for the observed result in any analytic epidemiological study include:

3.1. **Chance**

- . obtaining **adequate sample size** for the study could reduce the likelihood of chance as a possible explanation.
- . **statistically significant** finding leave little room for chance.

3.2. **Bias**

- . selection bias is best eliminated by **randomization**
- . information bias can be eliminated by:
 - . using **blinding** procedures
 - . using standard and comparable exposure and outcome ascertainment in both groups.

3.3. Confounding

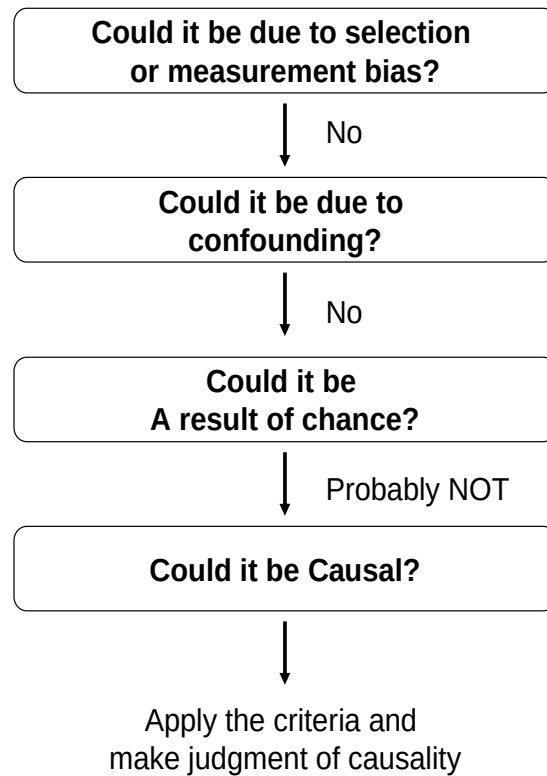
- . ways to control for confounding include:
- . use appropriate analytic tools to control known confounding factors - **multivariate analysis**
- . control for known and unknown confounders can be best achieved by **randomization**
- . matching if properly applied, is another method used for control of known confounders. Matching cases and controls on several characteristics creates difficulties in finding adequate number of control, require more complex statistical analysis, and run the risk of “overmatching”. Overmatching leads to missed associations due to inadvertent matching of causal factors.
- . compare basic socio-demographic characteristics to assure that balance was achieved.

Summary of the strengths and weaknesses of the common epidemiological study designs.

Theme	Cross-sectional	Cross-control	Cohort	Intervention/Trial
1.Ease	Difficulty depends on the study. Studies of natural living populations are hard compared with those at schools or other institutions	Usually difficult because of need for appropriate control group and problem of recall bias	Difficult because of added complexity of follow-up	Difficulty exceeds the cohort because of technical and ethical challenges of imposing an intervention
2 Timing	Usually finished within months or a few years	Usually finished within months or few years except those on incident cases of rare diseases	Usually long-term (decades) though sometimes (e.g. studies of birth outcomes) they can be quick	Usually deliberately designed
3 Maintenance and continuity	Study is usually stopped	Study is usually stopped	Long-term continuity is essential and problematic, particularly as observations are on free-living people	Similar to cohort studies but when trials are in patients with diseases, the commitment to the trial may be high
4 Costs	Costs depend on study but lower than cohort or trial of same size	Costs are usually comparable with cross-sectional studies and, as study size is small, the overall costs may be low	Costs are high both because numbers studied are large and because costs of retaining staff and system to collect data over many years are high	Costs are high for the same reason as the cohort study and there are additional costs of the intervention, obtaining ethical approval, and trial management
5 Ethics	Standard ethical issues and problem of obtaining access to sampling frame	Standard ethical issues as in clinical case-series but also those of cross-sectional studies for community controls	Confidentiality issues are acute, particularly as adverse outcomes may affect occupation are insurance premiums, potential intrusion of repeated contact and measurement	The ethics of trials are complex and evolving and hinge on the issue of doing no harm and informed consent
6 Data utilization	Usually under-utilized, as more information is collected than needed	As analysis is straight-forward, data are usually fully analyzed	Data tend to be underutilized	Data concerning the central questions are utilized
7 Main contribution	Major contribution to burden of disease, substantial contribution to analysis of associations and may conform or spark hypotheses	Major contribution to clinical knowledge, and sparking/testing causal hypotheses. Control group may supply burden of need data	Major contribution to both burden of disease (incidence) and causal analysis	Main contribution if to understanding of effectiveness of interventions, and indirectly to disease mechanisms
8 Observer bias	Small studies may be done one observer, but for most studies inter-observer bias is a problem	Small studies may be done by one observer; large studies usually need few	Usually requires multiple observers through exceptionally, studies may be small	Usually requires multiple observer
9 Selection bias	Selection bias arising from non-response is almost inevitable	Studies of prevalent cases have selection bias, those if incident cases minimize this. All studies have recall bias	Selection bias due to non-response at baseline is augmented by loss to follow-up	Selection biases particularly severe because non-participation may only be suitable for some of the target population
10 Analytic output	Main output is prevalence through other measures including the odds ratio are possible (not the relative risk)	Proportions exposed and odds ratios	Incidence rate and the relative incidence, i.e. relative risk	incidence, survival and numbers needed to treat or prevent

VI. EVALUATION OF EVIDENCE

Figure 6.1 Judging Observed Association



6.1. Accuracy of Measurement

$$\text{Accuracy} = \text{Validity} + \text{Precision}$$

Error is the difference between a computed or measured value and a true or theoretically correct value. Physical objects and parameters are accurately measured on an agreed up on measurement. For example, height is can be measured accurately using standardized height measurement. In health and disease the truth is usually unknown and can not be defined and computed. False knowledge generated from errors can be shown to wrong only with time and deeper study; wrong knowledge may persist unless disproved by another study. Error should be considered an inevitable and important part of epidemiological study. Bias is a more subtle matter than error and is a preference or an inclination, especially one that inhibits impartial judgment or that leads to an unfair act or policy stemming from prejudice. Error is common in science, as described by Popper (Popperian view) science progresses by the rejection of hypotheses (by falsification) rather than the establishment of so-called truths (by verification). Biological variations such that due to the circadian rhythms and limitations in measurement techniques due to technology, cost or ethical considerations makes error free measurements impossible.

Validity is the extent to which data collected actually reflect the truth. The concepts of **sensitivity** (ability to detect true positive) and **specificity** (ability to detect true negatives) can be used to characterize the validity of a measure ("measurement validity"). Study results are also described as "valid" when there is no systematic misrepresentation of effect or "bias" ("validity in the estimation of effect"). Validity is often described as internal or external.

Internal validity concerns the validity of inferences that do not proceed beyond the target population for the study. Internal validity is threatened when the investigator does not have sufficient data to control or rule out competing explanations for the results.

External validity, on the other hand, concerns generalizeability, or inferences to populations beyond the study's restricted interest. External validity is threatened, for example, when the investigator attempts to apply the findings of the study to a population which is not comparable to the population in which the research was completed. Internal validity should be the primary objective in study design, however, since efforts to

ensure the generalizeability of results often introduce problems of bias or confounding.

Precision, on the other hand, describes the extent to which random error (i.e., sampling variation and the statistical characteristics of the estimator) alters the measurement of effects. Misclassification may result in problems with either validity (due to systematic misclassification bias attributable to methodological aspects of study design or analysis) or precision (due to random misclassification error attributable to sampling variation). Random misclassification errors always bias measures of relative risk toward one. Systematic misclassification bias can either increase or decrease the strength of the measured association.

6.2. Bias

Bias may be defined as any systematic error in an epidemiologic study that results in an incorrect estimate of the association between exposure and risk of disease.

Bias may result from systematic error (or difference between exposed and unexposed populations or between cases and controls) in the collection, recording, analysis, or interpretation of data. Bias is an error that affects one group more than another. It could be intentional or unintentional. Evaluating the role of bias as an alternative explanation for an observed association is a necessary step in interpreting any study result. Unlike chance (including lack of precision) and confounding, which can be evaluated quantitatively, the effects of bias are far more difficult to evaluate and may even be impossible to take into account in the analysis. Bias results in false understanding about differences between groups and generates misleading patterns of health problems. For this reason, it is important to design and conduct studies in such a way that every possibility for introducing bias has been taken into account and to take steps to minimize chances of bias. In evaluation of study results, it is important to estimate the magnitude and direction of any suspected bias.

Types of bias may be grouped into two broad categories:

- 1) Selection bias refers to any error that arises in the process of identifying the study populations. Selection bias can occur whenever the identification of individual subjects for inclusion in the study on the basis of either exposure (cohort) or disease (case-control) status depends in some way on the other axis of interest.

Examples of selection bias include:

- 1) Berkson's bias - Case-control studies carried out exclusively in hospital settings are subject to selection bias attributable to the fact that risks of hospitalization can combine in patients who have more than one condition.
 - 2) Ascertainment bias - Differential surveillance or diagnosis of individuals make those exposed or those diseased systematically more or less likely to be enrolled in a study.
 - 3) Non-response bias - Rates of response to surveys and questionnaires in many studies may also be related to exposure status, so that bias is a reasonable alternative explanation for an observed association between exposure and disease.
 - 4) Loss to follow-up - This is a major source of bias in cohort studies. Persons lost to follow-up may differ from with respect to both exposure and outcome, biasing any observed association.
 - 5) Volunteer/Compliance bias - In studies comparing disease outcome in persons who volunteer or comply with medical treatment to those who do not, better results might be expected among those persons who volunteer or comply than among those who do not.
 - 6) Cohort bias - Refers to the biased view of the natural history of disease presented in survival cohorts, since only the prevalent cases (those with less lethal disease) are available for study in the latter part of the period of observation.
- 2) Observation or information bias includes any systematic error in the measurement of information on exposure or outcome.

Examples of Information bias include:

- 1) Interviewer bias - This can occur if the interviewer or examiner is aware of the disease status (in a case-control study) or the exposure status (in cohort and experimental studies). This kind of bias may affect every kind of epidemiologic study.

-
- 2) Recall bias - May result because affected persons may be more (or less) likely to recall an exposure that healthy subjects, or exposed persons more (or less) likely to report disease. This source of bias is more problematic in retrospective cohort or case-control studies.
 - 3) Social desirability bias - Occurs because subjects are systematically more likely to provide a socially acceptable response.
 - 4) Hawthorn effect - Refers to the changes in the dependent variable which may be due to the process of measurement or observation itself.
 - 5) Placebo effect - In experimental studies which are not placebo-controlled, observed changes may be ascribed to the positive effect of the subject's belief that the intervention will be beneficial.
 - 6) Regression to the mean - Refers to the statistical phenomenon that extreme values will tend to "regress" to more average values. Thus a change from a very high or very low values in the dependent variable may be attributable to simple random variation, rather than to changes in the independent variable.
 - 7) Healthy worker bias - Refers to the bias in occupational health studies which tend to underestimate the risk associated with an occupation due to the fact that employed people tend to be healthier than the general population.
 - 8) Lead-time bias - Results in overestimation of the effectiveness of a screening program for a condition which is actually caused by the early detection of a condition. It is more exaggerated in conditions with a long "lead-time" (such as cervical carcinoma). The detection of a condition before the person shows clinical signs and symptoms (the "lead time") is the cause of the measurement of prolonged survival in persons who participate in screening programs rather than a real prolongation of a real survival. Those individual who are diagnosed early may actually gained more "disease time".

- 9) Length/time bias - Occurs in studies of screening tests for cancer. This occurs due to the fact that screening tests for cancer tend to detect more slow-growing tumours with a better prognosis (since faster growing tumours are more often detected because they cause symptoms). As a result, the mortality rate of cancers found on screening will appear better than that of tumours not found on screening (though the effect is not due to the screening itself).

Some recommendations to minimize bias at the time of study design are:

- 1) Choose study design carefully. If ethical and feasible, a randomized double blind trial has the least potential for bias. If loss to follow-up will not be substantial, a prospective cohort study may have less bias than a case-control study. Controls for case-control studies should be maximally comparable to cases except for the variable under study.
- 2) Choose "hard" (i.e., objective) rather than subjective outcomes.
- 3) "blind" interviewers or examiners wherever possible.
- 4) Use well-defined criteria for identifying a "case" and use closed-ended questions whenever possible.
- 5) Collect data on variables you do not expect to differ between the two groups. If such a "dummy" variable regarding exposure, for example, in a case-control study shows an unexpected difference, it may alert you to recall bias.

6.3. Confounding

Confounding is the mixing of the effect of an extraneous variable with the effects of the exposure and disease of interest.

or

Confounding is the error in estimation of the measure of association between a specific risk factor and disease outcome, which arises when there are differences in the comparison populations other than the risk factor under study (Bhopal 2002).

or

Confounding is distortion of the estimated effect of an exposure on an outcome, caused by the presence of an extraneous factor associated both with the exposure and the outcome, i.e., confounding caused by a variable that is a risk factor for the outcome among non-exposed persons, and is associated with the exposure of interest, but is not an intermediate step in the causal pathway between exposure and outcome (Last 2001).

Confounding arises when some cause other than the exposure under study is more, or less, prevalent in the exposed group than in the unexposed. Such variable is defined as an extraneous (third) variable which is associated with the exposure and, independent of that exposure, be a risk factor for the disease. Confounding is a very difficult concept to understand quickly and students are advised to read through the note carefully and repeatedly. Confounding is a major cause of bias in epidemiology and aggravated by the failure to respect the cardinal rule 'compare like-with-like'- orange-with-orange, not orange-with-apple. However, except in experimental research this rule is rarely achieved in epidemiology. The most important analysis in all epidemiological studies is to compare the characteristics of the population under study with regard to the factors that are known or suspected to influence causation. Unknown factors are believed to be distributed equally between comparison groups if allocation is done randomly.

Characteristic of a confounding variable

1. Associated with the disease of interest in the absence of exposure
 - 1.a Risk factor for the study outcome among exposed group
 - 1.b Risk factor for the study outcome among non-exposed
2. Associated with the study exposure but not as a consequence of the exposure.

Effect of Confounding

Without prior knowledge of the effect of the variable on the outcome and exposure it is very difficult to predict the direction of effect of a suspected confounding variable. However, the effect could be categorized into three:

1. Totally or partially accounts for the apparent effect
2. Mask an underlying true association
3. Reverse the actual direction of the association

Control for Confounding Variables

The list of potential confounders in a study is limited to established risk factors for the disease of interest, though still some other variables may play a confounding role in the association it might be difficult to identify them and explain their effects.

In the design confounding could be minimized by:

Randomization
Restriction
Matching

Evaluation of confounding in the analysis by:

Standardization
Stratification/pooling
Multivariate analysis

6.4. Chance

One of the alternative explanations to the observed association between an exposure and a disease is chance. Since the general aim of epidemiological studies is to make generalization about a larger group of individuals on the basis of a sample population it is always important to evaluate the role of chance or sampling variability in any study which tries to elucidate association. Evaluation of the role of chance is mainly the domain of statistics and it involves:

6.4.1. Hypothesis Testing (Test of Statistical Significance)

Test of statistical significance quantifies the degree to which sampling variability may account for the observed results. The "P value" is used to indicate the probability or likelihood of obtaining a result at least as extreme as that observed in a study by chance alone, assuming that there is truly no association between exposure and outcome under consideration (i.e., H_0 is true). For medical research, the P value ≤ 0.05 is set conventionally to indicate statistical significance.

P value is a function of:

- the magnitude of the difference between the groups
- sample size

The fact implies that even a very small difference may be statistically significant if the sample size is sufficiently large, and a large difference may not achieve statistical significance if variability is substantial due to a small sample size. Hence, one cannot make a definite decision about the role of a factor based only on the P value.

Steps in testing for statistical significance

1. Assume that the exposure is not related to disease - state the null hypotheses.
2. Compute a measure of association - relative risk or odd ratio.
3. Calculate chi-square statistical test of significance.
4. For the value of chi-square calculated, look up its corresponding p-value in the table of chi-squares.

*** A very small p-value means that you are very unlikely to observe such an association if the null hypotheses is true.**

6.4.2. Estimation of Confidence Interval

The confidence interval represents the range within which the true magnitude of effect lies within a certain degree of assurance. It is more informative than just P value because it reflects on both the size of the sample and the magnitude of the effect.

6.5. Establishing a Causal Association

Cause and effect understanding is the highest form of achievement in scientific knowledge. Causal knowledge is the basis for rational actions to break the links between the factors causing the disease and disease itself. As in other sciences, epidemiological understanding of cause and effect does not have to be 100 percent complete or accurate to permit useful application, due ethical reasons even partial understanding must be applied quickly and effectively as possible for it may be a life and death matter. However, the application of incomplete knowledge requires more experience and cautionary application since it may have a devastating effects..

“To know the causes of disease and to understand the use of the various methods by which disease may be prevented amounts to the same thing as being able to cure the disease”- Hippocrates.

Sequence in establishing a causal association between an exposure and outcome”

1. Incidental observation of possible causal association between an exposure and outcome.
2. Descriptive epidemiologic analysis establishing the association on a population level.
3. Analytic epidemiologic studies establishing the association on an individual level.
4. Experimental reproduction of the outcome by the exposure and/or looking for biologic explanation.
5. Observation that removal of the exposure (or modification of the host response to it) decreases the occurrence of the outcome.

Our primary objective in epidemiology is to judge whether an association between exposure and disease is, in fact, causal. A cause in epidemiology can be defined as something that alters the frequency of disease, health

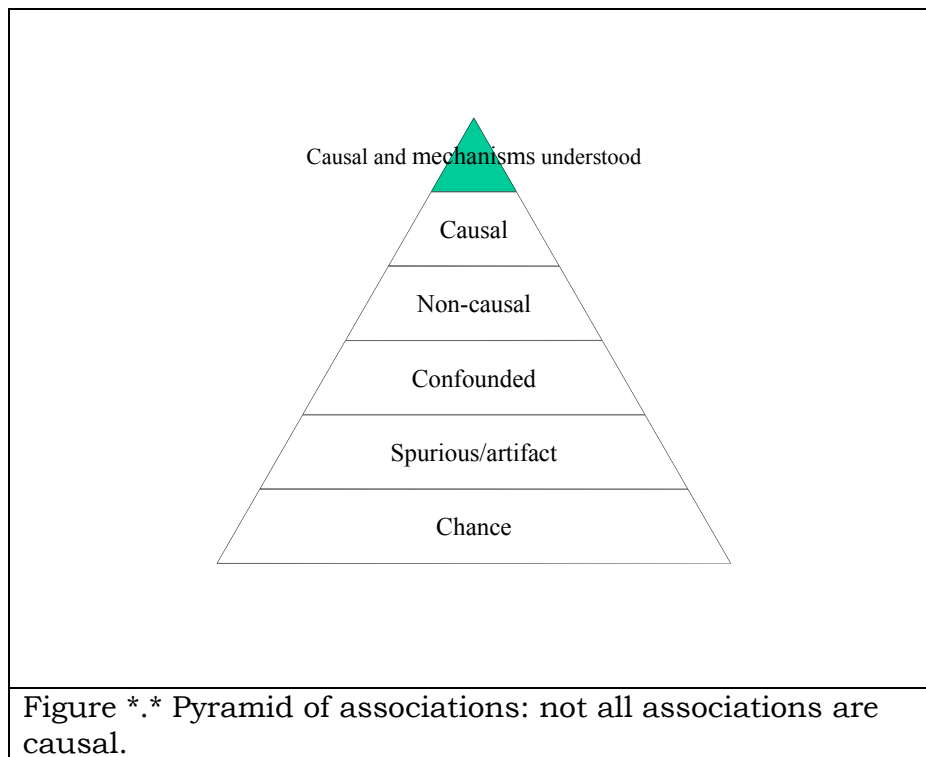
status, or associated factors in a population. Scientific proof of a cause-effect relationship is often difficult to obtain, since experimental studies are often neither feasible nor ethical. Since associations documented by other kinds of epidemiologic studies do not constitute proof of causation, one must assess the validity of individual studies and examine the totality of evidence from all available studies and make a judgement about the likelihood of a cause-effect relationship.

Judgements of causality must first consider whether, for any individual study, the observed association is valid (i.e., whether the findings reflect the true relationship between exposure and disease or may be explained by chance, bias, or confounding) and, second, whether the accumulated evidence supports a cause-effect relationship. The validity of an observed association is established by eliminating alternative explanations of that association. Associations can be:

1. Artefactual (spurious) associations, which may be:
 - a) **The result of chance variation (i.e., type 1 error).** Statistical tests and confidence intervals can help to evaluate the likelihood of this as an explanation for an association.
 - b) **The result of bias**, or systematic error in the design or conduct of the study. Examples of how bias can lead to artefactual (i.e., not real) associations are presented below.
2. Non-causal (indirect) associations, which may occur when:
 - a. the associated factor is itself an effect, rather than a cause (**reverse causation**), or both a cause and an effect (**reciprocal causation**). For example, in the association between vitamin A deficiency and diarrhoea, vitamin A deficiency could be a cause or an effect of diarrhoea, or both. Vitamin A deficiency results in abnormalities in epithelial surfaces and, thus could impair resistance to the infectious agents of diarrhoea. On the other hand, diarrhoea leads to reduced food intake through loss of appetite and to impaired absorption of nutrients, both of which could result in vitamin A deficiency after repeated episodes of diarrhoea.
 - b) **The association is due to a confounding effect by a third variable.** A confounding variable is one independently associated with both the exposure and the disease. To confound, a variable must fulfil each of the following two criteria: 1) it must be related to both the frequency of disease exposure and the frequency of disease recognition, and 2) it must occur with differing frequencies in groups being compared (cohorts or cases and controls). For example, the association between anaemia and illiteracy is likely non-causal, due rather

to the confounding effect of socioeconomic status, which is independently related to both anaemia (because of poor diet) and illiteracy (because of reduced access to educational opportunities).

3. Causal associations, which can be established only when other potential explanations of the association can be ruled out.
- 4.



In observational studies, there are many potential confounders and sources of bias, some of which may remain undetected. The results of one observational study rarely provide adequate support for concluding that there is a cause-and-effect relationship between an exposure and a disease. Properly conducted experimental trials do provide direct proof of causality, yet are often impossible because of ethical considerations.

In the absence of experimental evidence, the following criteria (called the **Bradford-Hill criteria**) are used to assess the strength of evidence for a cause-and-effect relationship. The criteria are listed in descending order of importance (see also Table *.* for more detail implications of each criterion):

1. **Strength of the Association** - The stronger the association, the more likely that it is causal.
2. **Consistency of the Relationship** - The same association should be demonstrable in studies with different methods, conducted by different investigators, and in different populations.

3. **Specificity of the Association** - The association is more likely causal if a single exposure is linked to a single disease.
4. **Temporal Relationship** - The exposure to the factor must precede the onset of the disease.
5. **Dose-response Relationship** - The risk of disease often increases with increasing exposure to a causal agent.
6. **Experimental confirmation**- Confirmation that the risk of disease often increases with increasing exposure to a causal agent by manipulating exposure level.
7. **Biological Plausibility** - The hypothesis for causation should be coherent with what is known about the biology and the descriptive epidemiology of the disease.

Illustration of Using Causation Judgement Decisions

Questions underlying criterion	Label of criterion	Evidence on criterion:		
		Unsure	No	Yes
Does exposure to the cause raise the incidence of the disease?	Strength of association	Judgment premature	Not causal	Causal relationship possible
Is the association consistent across different studies and between subgroups?	Consistency	Defer decision and await further research	Judgement will require explanation of inconsistent results	Strengthens casual claim
Is the effect of risk factor(s) limited to the particular disease of interest?	Specificity	Not critical	Not critical but added caution	Strengths causal claim
Does the supposed cause precede the disease?	Temporality	Judgement premature	Not causal	Causal relation possible
Does varying exposure lead to varying amounts of disease?	Dose response	Not critical	Causal relation still possible if there is a threshold effect	Strengthens case for a causal claim
Does manipulating the level of exposure change the level of the disease?	Experimental confirmation	Not always possible, so not critical	Caution needed for a causal claim	Strong confirmation of a causal relation
Is the way that the cause exerts its effect on disease understood?	Biological plausibility	Not critical	Not critical but great caution needed for causal claim	Causal judgment strengthened

VII. PRESENTATION OF EPIDEMIOLOGIC INFORMATION

The ultimate goal of epidemiologic studies is to generate information which are useful for planning of health services and promotive, preventive and control activities. Usually these studies generate an enormous data which are difficult to comprehend in the form they are collected. Therefore, it is mandatory to reduce the data in the form easily understandable by everybody. To that end epidemiologist use several data reduction/summary methods to display their data as simple as possible. Some of the methods are discussed below.

7.1. Table

Table summarize a set of data arranged in rows and columns. Tables are useful for demonstrating patterns, exceptions, differences or other relationships. Tables may also serve as the basis for preparing more visual displays of data, such as graphs and charts, where some of the detail may be lost. Tables designed to present data should be as simple as possible. Two or three small tables, each focusing on a different aspect of the data, are easier to understand than a single large table that contains many details or variables. To create a table that is self-explanatory, use the following guidelines:

- Use a clear and concise title that describes the what, where, and when of the data in the table. Precede the title with a table number.
- Label each row and each column clearly and concisely and include the units of measurement for the data.
- Show totals for rows and columns. If you show percents, also give their total (always 100).
- Explain any codes, abbreviations, or symbols in a footnote.
- Note any exclusion in a footnote.
- Note the source of the data in a footnote if the data are not original.

Types of tables

7.1.1. One-variable table(frequency distribution table)

This displays the values or categories of one variable and the number and percentage of people falling into that category.

Table 7.1: Distribution of students by sex.

Category	Number	Percent
Female	80	40
Male	120	60
Total	200	100

7.1.2. Two-variable table (contingency table)

Data displayed in contingency tables is frequently used to calculate measures of association and tests for statistical significance. The cells of a table can just as easily contain means, rates, years of potential life lost, relative risks, and other statistical measures. A common type of contingency table is the two-by-two table, in which each of the two variables has only two categories.

Table 7.2: A two-by-two table of measles and vaccination.

		Measles	
		Yes	No
Vaccination	Yes	10	90
	No	70	30

7.1.3. Three-variable table

Though its use is not much recommended sometimes three variables can be displayed in a table. At this point it is important to remember that elegant tables are simple and easy to understand.

7.1.4. Table shells or dummy tables

Dummy tables are prepared as part of the analysis plan to show how the data will be organised and displayed once the data is collected. Table shells are complete except for the data, showing titles, headings and categories. In developing table shells which include continuous variables such as age, we create more categories than we may later use, in order to disclose any interesting patterns.

Table 7.3: Dummy Table for Distribution of children in village X by age.

Category(in year)	Number	Percent
<1		
1-4		
5-9		
10-14		
Total		

Ordinal variables are presented according to their intrinsic natural categories. For continuous variables artificial categories must be created based on the purpose of the study, however, it is advisable to create more categories (narrow intervals) in order not to miss any interesting patterns.

In creating categories, or class intervals, for continuous variable of epidemiologic data, remember the following guidelines:

- Create class intervals that are mutually exclusive and that include all of the data.
- Use a relatively large number of narrow class intervals for your initial analysis. You can always combine intervals later. In general, you will have 4 to 8 intervals when analysis is complete.
- Use natural or biologically meaningful intervals when possible. Try to use age groupings that are standard or used most frequently in the particular field of study. If rates are to be calculated, the intervals for the numerator must be the same as the intervals used for the available population data.
- Create a category for unknowns.

Table 7. 4 Lists of some standard class intervals used for age (age-groupings) used for data presentation and analysis:

Table 4. SOME STANDARD AGE GROUPINGS FOR EPIDEMIOLOGIC REPORTING

<i>Notifiable Diseases</i>	<i>Pneumonia and Influenza Mortality</i>	<i>Final Mortality Statistics</i>	<i>HIV/AIDS</i>
<1 year	<28 days	<1 year	<5 years
1 -4	28 days-<1 year	1-4	5-12
5 -9	1-14	5-14	13-19
10-14	15-24	15-24	20-24
15-19	25-44	25-34	25-29
20-24	45-64	35-44	30-34
25-29	65-74	45-54	35-39
30-39	□ 85	55-64	40-44
40-49	unknown	65-74	45-49
50-59		75-84	50-54
□ 60		□ 85	55-59
age not stated		Not stated	60-64
			□ 65

If no natural or standard class intervals are available, several strategies can be used for creating intervals. These include:

- Divide the data into groups of similar size. To apply this strategy, divide the total number of observations by the number of intervals you wish to create (usually 4, but you might start with 8). Next, develop a cumulative frequency column of a rank-ordered distribution of your data to find where each interval break would fall.

-
- Base intervals on mean and standard deviation. With this strategy you can create 3, 4 or 6 class intervals.
 - Divide the range into equal class intervals. This method is most common and simplest, and is most readily adapted to graphs.

7.2. Graph

A graph is a way to show quantitative data visually, using a system of coordinates. It is a kind of statistical snapshot that helps us see patterns, trends, aberrations, similarities, and differences in the data.

We usually use the horizontal axis (or *x-axis*) to show the values of the independent (or *x*) variable. We use the vertical axis (or *y-axis*) to show the dependent (or *y*) variable, which is usually a frequency measure such as number of cases or rates of disease. Each axis should be labelled (with both the name of the variable and the units in which it is measured) and a scale of measurement marked along the line.

7.2.1 Histogram

A histogram is a graph of the frequency distribution of a continuous variable. It uses adjoining columns to represent the number of observations for each class interval in the distribution. The area of each column is proportional to the number of observations in that interval. Histograms with unequal class intervals are, therefore, difficult to construct and are not recommended. A second variable may be displayed using a histogram by shading each column into the component categories of the second variable. Epidemic curves (which are not really "curves" at all) are frequently displayed as histograms.

7.2.2 Frequency polygon

A frequency polygon, like a histogram, is the graph of a frequency distribution. In a frequency polygon, we mark the number of observations within an interval with a single point placed at the mid-point of that interval, and then connect the points with a straight line. A frequency polygon of a set of data must enclose the same area as a histogram of the data. (Note that for each area of histogram that the polygon leaves out, it includes another area of equal size.) Frequency polygons are often used to compare two or more distributions on the same axis. Other commonly used graphic displays of epidemiologic data include cumulative frequency curves, survival curves, and scatter diagrams (or scatter grams).

7.3. Chart

Charts are methods of illustrating statistical information using only one coordinate. They are most appropriate for comparing data with discrete categories other than "place". Variables shown in bar charts are either discrete or non-continuous (e.g., race or sex) or are treated as though they were discrete and non-continuous (e.g., age groups rather than age intervals). The length or height (bar charts can be presented either horizontally or vertically) of each bar is proportional to the frequency of the event in that category (and, therefore, scale breaks should not be used). The simplest bar chart is that used to display data from a one-variable table. This presentation makes it very easy to see the relative importance of different variables.

7.3.1 Grouped bar chart

A grouped bar chart can be used to illustrate data from two-variable or three-variable tables, when an outcome has only two separate categories. Bars within a group are usually adjoining, should be no more than three, and must be illustrated distinctively and described in a legend.

7.3.2 Stacked bar chart

Stacked bar charts can be used to show categories of a second variable as components of the bars that represent the first variable.

7.3.3 Deviation bar chart

Deviation bar charts can be used to show deviations in a variable, both positive and negative, from a variable.

7.3.4 100% component bar chart

100% component bar charts are useful for comparing the contribution of different components to each of the categories of the main variable. This is a variation of the stacked bar chart in which we make all the bars the same height (or length) and show the components as percents of the total rather than as actual values.

7.3.5 Pie chart

Pie charts are simple, easily understood charts in which the size of the “slices” shows the proportional contribution of each component part. Pie charts are useful for showing the component parts of a single group or variable. Conventionally, we begin at 12 o'clock and arrange the component slices from largest to smallest.

3.6 Geographic coordinate charts

Geographic coordinate charts (maps) are used to show the location of events or attributes. Spot maps use dots or other symbols to show where an event occurred or a condition exists.

Although it can show the geographic distribution of an event, a spot map does not show **risk** because it does not take the size of the population into account. Area maps can overcome this problem by using shaded or coded areas to show either the incidence of an event in sub-areas, or the distribution of some condition over a geographic area. Area maps can show either numbers or rates.

To construct a bar chart, observe the following guidelines:

- Arrange the categories that define the bars, or groups of bars, in a natural order, such as alphabetical or by increasing age, or in an order that will produce increasing or decreasing bar lengths.
- Position the bars either vertically or horizontally, except for

deviation bar charts, in which the bars are usually positioned horizontally.

- Make all of the bars the same width.
- Make the length of bars in proportion to the frequency of the event. Do not use scale breaks.
- Show no more than three bars within a group of bars.
- Leave a space between adjacent groups of bars, but not between bars within a group.
- Code different variables by differences in bar colour, shading, cross-hatching, etc. and include a legend that interprets your code.

Table 7.5

Guide to Selecting a Graph or Chart to Illustrate Epidemiologic Data

Type of Graph or Chart	When to Use
Arithmetic-Scale Line Graph	Trends in numbers or rates over time
Semi-logarithmic-scale line graph	1) Emphasize rate of change over time 2) Display values ranging over more than 2 orders of magnitude
Histogram	1) Frequency distribution of continuous variable 2) Number of cases during epidemic (epidemic curve) over time
Frequency Polygon	Frequency distribution of continuous variable, especially to show components
Cumulative Frequency	Cumulative frequency for continuous variables
Scatter Diagram	Plot association between two variables
Simple Bar Chart	Compare size or frequency of different categories of a single variable
Grouped Bar Chart	Compare size or frequency of different categories of 2-4 series of data
Stacked Bar Chart	Compare totals and illustrate component parts of the total among different groups
Deviation Bar Chart	Illustrate differences, both positive and negative, from baseline
100% Component Bar Chart	Compare how components contribute to the whole in different groups
Pie Chart	Show components of a whole
Spot Map	Show location of cases or events
Area Map	Display cases/events or rates geographically

VIII. OUTBREAK INVESTIGATION AND MANAGEMENT

Outbreak investigations are important and challenging component of epidemiology and public health. Properly conducted investigations can help identify the source of ongoing outbreaks and prevent additional cases. Even when an outbreak is over, which is the case in many developing countries; a thorough epidemiologic and environmental investigation often can provide a useful knowledge about the disease and prevent future outbreaks.

Outbreak investigations are, in theory, indistinguishable from other epidemiologic investigations (researches); however, outbreak investigations encounter more constraints. The following are some of major constraints:

- 1) If the outbreak is ongoing at the time of the investigation, there is great urgency to find the source and prevent additional cases.
- 2) Because outbreak investigations frequently are public, there is substantial pressure to conclude them rapidly, particularly if the outbreak is ongoing.
- 3) In many outbreaks, the number of cases available for study is limited; therefore, the statistical power of the investigation is limited.
- 4) Early media reports concerning the outbreak may bias the responses of persons subsequently interviewed.
- 5) Because of legal liability and the financial interests of persons and institutions involved, there is pressure to conclude the investigation quickly, which may lead to hasty decisions regarding the source of the outbreak.
- 6) If detection of the outbreak is delayed, useful clinical and environmental samples may be very difficult or impossible to obtain.

8.1 PATTERNS OF EPIDEMICS

Two principal types are well recognized. These are the common source and propagated/progressive. The two types can be distinguished by plotting an epidemic curve. An epidemic which shows the features of both types is referred as mixed.

8.1.1. Common source epidemics

Common source epidemics are caused by exposure of a group of people to a common noxious influence, such as an infectious agent or a toxin. If the exposure is brief and simultaneous all exposed will develop the disease within one incubation period - referred as point, or point source epidemic/outbreak (Figure 8.1). A rapid rise and fall of an epidemic curve suggests a point source epidemic- this is called log-normal distribution. If the source of an outbreak remains for a longer time, days, weeks or longer either continuously or

intermittently, there will be multiple exposures with variable incubation period, this will make an epidemic curve with no clear peak and the duration of the outbreak will be prolonged. Continuous common source - makes wide peak in the epidemic curve, because of the range of exposures and range of incubation periods. Intermittent common source - results in an irregular pattern of the epidemic curve that reflects the intermittent nature of the exposure.

Epidemic Curve of Point Source Epidemic

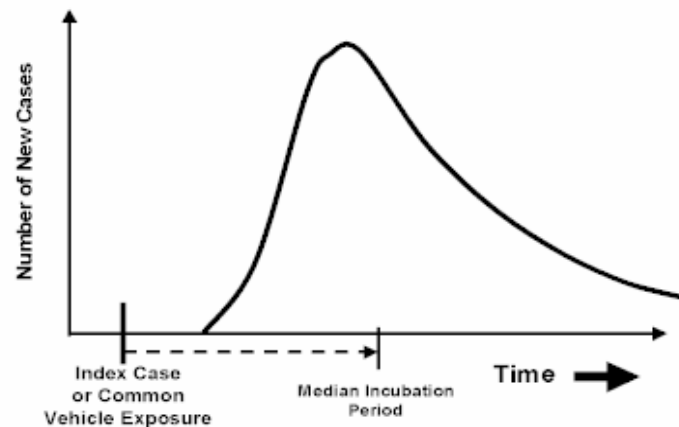


Figure 8.1 Epidemic Curve of Point Source Epidemic

8.1.2. Propagated or progressive epidemics

Outbreak of this type can occur through direct person-to-person transmission or the transmission could pass through a vector from infected to healthy person.

The epidemic curve would have a successive series of peaks reflecting increasing numbers of cases in each generation (Figure 8.2). The epidemic usually wanes after a few generations, either because the number of susceptible falls below some critical level, or because intervention measures become effective. In reality, few propagated outbreaks provide a classic pattern. Diseases with short incubation period and are highly infectious, can create a rapidly rising and falling epidemic curve similar to that of a point source epidemic.

Exposure of susceptible persons occurs over time rather than all at once as above. These epidemics progress through a group over a period of time that is considerably longer than the typical incubation period.

- When one can not distinguish the two by the epidemic curve, studying the geographic distribution will help to differentiate

them. The propagated epidemics tend to show geographic spread with successive generations of cases.

Epidemic Curve of Propagating Epidemic

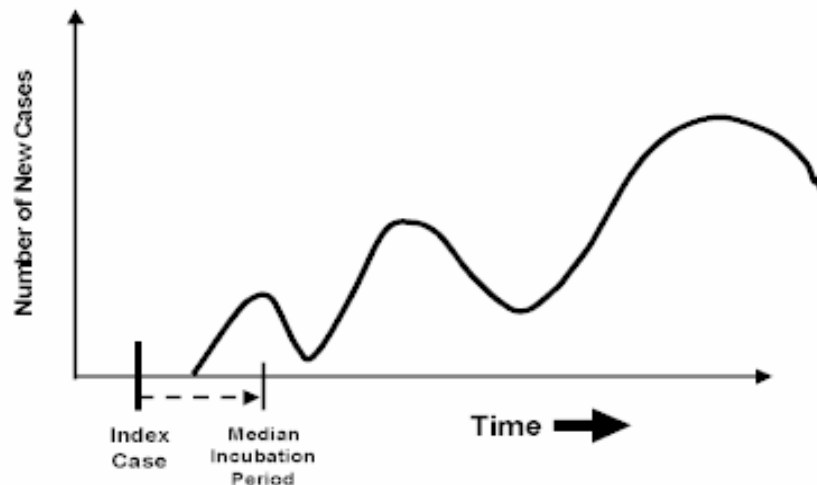


Figure 8.2 Epidemic Curve of Propagating Epidemic

8.1.3. Mixed Epidemics

Epidemics having the features of both common source and propagated epidemics are referred as mixed epidemics. For example a common source outbreak may be followed by secondary person-to-person spread.

8.2. Steps of an Epidemic Investigation

There is no rigid step to follow during investigation of an outbreak. Several activities could be accomplished simultaneously. The steps to follow are set by the individual investigator depending on the suspected cause of the outbreak. Verification of the diagnosis and establishment of the existence of an epidemic are commonly among the first steps.

8.2.1. Prepare for field work

Before leaving for the field an investigator must be well prepared to under take the investigation. Preparations can be categorized into three:

- A. Investigation related: Investigator must have the appropriate scientific knowledge, supplies, and equipment to carry out the investigation. Discuss the situation with knowledgeable people, review applicable literature, and collect sample questionnaire.
- B. Administration related: No matter there is urgency in handling the situation, it is useful to observe all administrative procedures. This include arrangement of transportation and organising personnel matters.
- C. Consultation: clarify your and your team role in the field. Identify local contacts at the site where the outbreak is reported and arrange where and when to meet them.

8.2.2. Verify the existence of an epidemic

Compare the current number of cases (or incidence) with the past levels of disease in that community, considering the seasonal variation in the occurrence of the disease, to determine whether an excessive number of cases have occurred, i.e., compare the observed number of cases (reported as outbreak) with the expected number of cases in the area.

Be careful, excess may not always indicate an outbreak. The excess may be due to changes in local reporting procedures, change in case definition, increased interest because of local or national awareness, or improvements in diagnostic procedures. In areas with sudden changes in population size such as resort areas, college towns, and migrant farming areas, changes in the numerator (number of reported cases) may simply reflect changes in the denominator (size of the population) - absolute numbers (without proportion or rates) should be carefully analyzed.

Outbreak/epidemic: is the occurrence of more cases of disease than expected in a given area or among a specific group of people over a particular period of time.

Cluster: is an aggregation of cases in a given area over a particular period without regard to whether the number of cases is more than expected.

8.2.3. Verify the diagnosis

Review the clinical and laboratory findings of the cases to establish the diagnosis. This is to ensure that the problem has been properly diagnosed and to rule out laboratory error as the basis for the increase in diagnosis. If you have any doubt about the laboratory findings review the laboratory techniques being used with the qualified laboratorian or send specimen for confirmation to reference laboratory.

Summarize the clinical findings with frequency distribution. They are useful in characterizing the spectrum of the illness, verifying the diagnosis, and developing case definitions. Visit as much patients as you can. Conversation with patients is very helpful in generating hypothesis about disease aetiology and spread. Depending on the type of the problem under investigation establish criteria for labelling persons as "cases".

8.2.3.1 Case definition

Case definition is a standard set of criteria for deciding whether an individual should be classified as having the health condition of interest.

A case definition includes clinical criteria, particularly in an outbreak investigation, restricted by time, place, and person. Set simple and objective measures, and apply them consistently and without bias to all persons under investigation. Do not include an exposure or risk factor which is going to be tested in the case definition. For example, if one of the goal of the investigation is to determine whether work place is associated with the illness, the case definition should not be restricted with regard to working place. Use "loose" case definition early in the investigation to identify the extent of the problem and the population affected. But, during testing the hypothesis generated from this process using analytic epidemiology, specific or "tight" case definitions must be used.

Keeping in mind the uncertainty of some diagnosis, it is advisable to classify cases as confirmed, probable, or possible. This may help to keep track of a case if the diagnosis is not confirmed or if there is a decision not to order laboratory test required to confirm the diagnosis

because the test is expensive, difficult to obtain, or unnecessary. It is also customary that investigators usually confirm the diagnosis of a few cases and rely on clinical features to identify the rest of the cases.

Confirmed/definite: a case with laboratory verification.

Probable : a case with typical clinical features of the disease without laboratory confirmation.

Possible : a case presented with fewer of the typical clinical features.

8.2.3.2 **Surveillance** - identifying and counting cases.

Often the cases which create the concern are small and non-representative fraction of the total number of cases. Therefore, epidemic investigators should "cast the net wide" to determine the geographic extent of the problem and the population affected by it. In order to do that one must adopt appropriate methods, for the setting and disease in question, to identify cases. The two types of surveillance commonly utilized in an outbreak investigation are:

1. Stimulated or enhanced passive surveillance includes:

- Sending out a letter describing the situation and asking for reports.
- Alerting the public directly, usually through local media, to see a physician if they have symptoms compatible with the disease in question.
- Asking case-patients if they know anyone else with the same condition.

2. Active surveillance:

- Making telephone call or visit the facilities to collect information on cases.
- Conducting a survey of the entire population.

Regardless of the disease under investigation collect the following types of information about every case:

- ☐ Identifying information - allows you to contact patients and to map the geographic extent of the problem.
- ☐ Demographic information - provides the "person" characteristics of the population at risk.
- ☐ Clinical information - allows verification of the case definition and charting the time course of the outbreak, and helps to describe the spectrum of the illness.
- ☐ Risk factor information - inquire specifically exposure to the suspected cause.

- Reporter information - to help you inquire additional information if there is a need or to report back the results of your investigation.

8.2.4. Describe the epidemic with respect to time, place, person.

Collect relevant information related to the investigation. Information could be obtained from the already existing records or you can obtain using a case investigation form specifically designed for a particular situation under investigation. Information must be collected carefully, so that, at the end they will enable the investigator to characterize the outbreak with respect to time, place and person.

By using well established descriptive epidemiological tools, such as the epidemic curve and spot mapping, an outbreak can be characterized by time, place and person.

Epidemic curve- plots the cases by the time of onset and provides a time frame for the outbreak investigation.

Spot map- plots the cases by location and shows the geographic spread of cases.

Attack rates- Calculate rates of illness in population at risk by exposure to specific suspected items and other relevant attributes. The identification of "relevant" attributes may be a crucial step in the solution of the problem.

8.2.5. Formulate and Test Hypotheses

Formulate the hypotheses based on your characterization of the epidemic by time, place, and person. The hypotheses should address the source of the agent, the mode of transmission, and the exposures that caused the disease. Determine the type of epidemic- common source Vs propagated. Based on characteristics of the epidemic define the population at the highest risk and consider the possible source(s) of the disease (infection). The hypotheses should be testable.

In an outbreak investigation, evaluation of hypotheses can be done in two ways: either by comparing the hypotheses with the established fact, or by using analytic epidemiology to quantify relationships and explore the role of chance.

*** Analytic approach:**

The analytic technique utilizes the cohort and the case-control approach to identify possible source of an outbreak. Cohort approach identifies the comparison group based on exposure status. The case-control method identifies the comparison groups on the basis of their disease status.

- Compute Odds ratio to find association between cases and controls (non-ill) with regard to exposure to the suspected cause - case-control.
- Calculate relative risk (attack rate) to determine whether there is association between exposed and non-exposed - cohort.
- Compute statistical tests to determine how likely it is that the investigation results could have occurred by chance alone, if exposure was not actually related to disease.

In both analytic approaches a test of significance has to be worked out to determine whether the differences observed between the two groups (cases Vs control, or exposed Vs non-exposed) are not due to chance. Statistically significant difference only provides supportive evidence on the possible source of an outbreak. Causation can only be established after careful assessment of the whole situation and requires laboratory proof, which is not always easy.

$$\begin{aligned}
 \text{Relative Risk} &= \frac{\text{Incidence (attack) rate among exposed}}{\text{Incidence (attack) rate among unexposed}} \\
 &= \frac{a/a + b}{c/c + d} \\
 \text{Odds Ratio} &= \frac{\text{Proportion of exposed in diseased (cases)}}{\text{Proportion of exposed in non-diseased (controls)}} \\
 &= \frac{a/a + c}{b/b + d}, \text{ if } a \text{ \& } b \text{ are small relative to } c \text{ \& } d \\
 &= \frac{a/c}{b/d} = \frac{ad}{bc}
 \end{aligned}$$

8.2.6. Search for additional cases: Locate unrecognized or unreported cases.

Passive: inquire physicians or hospitals or both whether they have seen similar cases.

Active: do intensive investigation in the community on asymptomatic persons or contacts of the cases. For example: doing liver function test in an investigation of hepatitis A outbreak.

8.2.7. Analyze the Data

- . Assemble all results.
- . Interpret findings.

8.2.8. Make a decision on the hypotheses tested.

- . All the findings must be consistent with one, and only one, hypotheses.

8.2.9. Intervention and follow-up.

Although it is discussed late, intervention must start as soon as possible depending on the specific circumstances. Aim control measures at the weak link or links in the chain of infection. One might aim control measures at the specific agent, source, or reservoir. For example, an outbreak might be controlled by destroying contaminated foods, sterilizing contaminated water, or destroying mosquito breeding sites. or an infectious food handler could be removed from the job and treated. (See discussion on epidemic management).

8.2.10. Report of the investigation

At the end prepare a comprehensive report and submit to the appropriate/concerned agency (or agencies). The report should follow the usual scientific format: introduction, background, methods, results, discussion, and recommendations.

The report should discuss in detail:

- factors leading to the epidemic.
- evaluation of measures used for the control of the epidemic.
- recommendations for the prevention of similar episodes in the future.

8.3. Managing Outbreak/epidemics

Management of epidemics require an urgent and intelligent use of appropriate measures against the spread of the disease. Action to be taken is dependent on the type of the disease as well as the source of the outbreak. However, the action can be generally categorized as presented below to facilitate easy understanding of the strategies.

8.3.1. Measures Directed Against the Reservoir

Understanding the nature of the reservoir is necessary in the selection of an appropriate control methods and their likelihood of success. The following are examples of control measures against disease with varies reservoir :

Domestic animals as reservoir:

- . Immunization
 - . testing of herds
 - . destruction of infected animals
- Example : brucellosis and bovine tuberculosis.

Wild animals as reservoir:

- . post-exposure prophylaxis
- Example : rabies

Humans as reservoir

- removal of the focus of infection- e.g., cholecystectomy in a chronic typhoid carrier.
- Isolation of infected persons. This is separation of infected persons from non-infected for the period of communicability. Not suitable in the control of diseases in which a large proportion are inapparent infection or in which maximal infectivity precedes overt illness.
- Treatment to make them non-infectious: e.g., tuberculosis.
- Disinfection of contaminated objects.

□ Quarantine- is the limitation of freedom of movement of apparently healthy persons or animals who have been exposed to a case of infectious disease. Usually imposed for the duration of the maximal incubation period of the disease.

* Cholera, Plaque, and yellow fever are the three internationally quarantable diseases by international agreement.

* Now quarantine is replaced in some countries by active surveillance of the individuals - maintaining close supervision over possible contacts of ill persons to detect infection or illness promptly; their freedom of movement is not restricted.

8.3.2. Measures that interrupt the transmission of organisms

- * Action to prevent transmission of disease by ingestion:
 - . purification of water
 - . pasteurization of milk
 - . inspection procedures designed to ensure safe food supply.
 - . improve housing conditions.
- * Attempts to reduce transmission of respiratory infections
 - . chemical disinfection of air and use of ultraviolet light.
 - . work on ventilation patterns, like unidirectional ("laminar") air flow to reduce the transmission of organisms in hospitals.
- * Action to interrupt transmission of diseases whose cycles involve an intermediate host
 - . clearing irrigation farms from snails to control schistosomiasis.

8.3.3. Measures that reduce host susceptibility

- * Active immunization, when either the altered organism or its product is given to a person to induce production of antibodies - EPI.
- * Passive immunization, has lesser role in the control of communicable diseases than active immunization:
 - Transfer of maternal antibodies to the fetus through the placenta.
 - Prophylaxis administration of immune serum globulin (ISG). E.g., TAT for un-immunized persons who receive penetrating wounds, antitoxin against *Clostridium botulinum* and antiserum against rabies.
 - Chemoprophylaxis:
 - use of antibiotics for known contacts of cases- for example, in tuberculosis, gonorrhoea, and syphilis.
 - use of chlorquine to persons travelling to malaria endemic areas.

8.4. Uncovering outbreaks

Outbreaks are detected in one of the following ways:

- . Through timely analysis of routine surveillance data, this may reveal an increase in reported cases or unusual clustering of cases.
- . Report from clinician.
- . Report from the community, either from the affected group or concerned citizen.

8.5. Why Investigate Possible Outbreaks

1. To institute control and prevention measures

In order to design and implement appropriate control measures assessment of the extent of the outbreak and the size and the characteristics of the population at risk needs to be done.

2. Opportunity for research

Outbreaks are natural experiment waiting to be analyzed and exploited. It gives a unique opportunity to study the natural history of diseases. It may also help to assess the impact of control measures and the usefulness of new epidemiology and laboratory techniques.

3. Training opportunity

Investigating an outbreak requires a combination of diplomacy, logical thinking, problem-solving ability, quantitative skills, epidemiologic know-how, and judgement. These skills improve with practice and experience. Therefore, an outbreak may provide a good opportunity for an epidemiologist in-training to learn these skills by working with experienced epidemiologist.

4. Opportunity for program evaluation

An outbreak of a disease targeted by a public health program, such as EPI, tuberculosis or STDs, may reveal a weak point in that program and provide the opportunity to change or strengthen the program's effort.

5. Public, political, or legal concerns

Public, political, or legal concerns sometimes override scientific concerns in the decision to conduct an investigation. The call from these parties usually has no scientific basis and such investigations mostly do not identify a causal link between exposure and disease. Nevertheless, health departments have to be responsive to public concerns, because it at least provides an opportunity to educate the public.

IX. EPIDEMIOLOGIC SURVEILLANCE

Epidemiologic Surveillance is the systematic collection, analysis, interpretation and dissemination of health data in an ongoing basis. Surveillance provides "information for action" which can be used to investigate, prevent, and control disease in communities. Its purpose is to provide a factual basis for setting priorities, planning programs, and taking action to promote and protect community health. Surveillance can be conducted globally (as in the AIDS surveillance system managed by WHO), regional (as in the polio surveillance in Latin America), national, or institutional (as in the surveillance for hospital acquired or nosocomial, infections or for potential causes of epidemics in refugee camps).

We do not limit surveillance to diseases for which we have effective control measures. Surveillance can be justified for two additional purposes: 1) to learn more about the natural history, clinical spectrum, and epidemiology of a disease, and 2) to obtain baseline data which we can use to assess the effectiveness of prevention and control measures when they are developed and implemented.

Surveillance is a system of close observation of all aspects of the occurrence and distribution of a given disease through systematic collection, tabulation, analysis, and dissemination of all relevant data pertaining to that disease.

We monitor health events for the following purposes:

- To detect sudden changes in disease occurrence and distribution (determines the need for epidemic investigation and control) and to ensure that effective action to control the disease is being done.
- To follow secular (long-term) trends and patterns of disease (alerts decision makers of the need to reallocate resources or shift policy)
- To identify changes in agents and host factors (helps to assess the potential for future disease occurrence)
- To detect changes in health care practices (points up the need for changes in preventive measures)

Interpretation of surveillance data may also provide the basis for generating hypotheses and stimulating community health research, test hypotheses regarding the impact of exposures on disease occurrence. Archival surveillance data have also been used to develop statistical models of diseases, such as to predict the feasibility of proposed programs to eradicate measles and polio.

The following are some key sources of surveillance data, not all of which are available in every country:

- Census data
- Mortality reports (birth and death certificates, autopsy reports)
- Morbidity reports (notifiable disease reports)
- Hospital data (discharge diagnoses, surgical logs, hospital infection reports)
- Absenteeism records (school, workplace, compensation claims)
- Epidemic reports
- Laboratory test utilization and result reports
- Drug utilization records
- Adverse drug reaction reports
- Special surveys (e.g., research data, serologic surveys)
- Police records (especially for injury, alcohol-related crime)
- Information on animal reservoirs and vectors (e.g., for rabies, plague, Lyme disease)
- Environmental data (hazard surveillance, water and food testing)
- Special surveillance systems (e.g., for injury and occupational illness)

9.1. Types of Surveillance

Passive surveillance is that in which health care providers send reports based on a known set of rules and regulations.

Active surveillance is that in which public health officials contact providers to solicit reports of events or diseases. Such active surveillance is usually limited to specific diseases over a limited period of time, such as after a community exposure or during an epidemic. Incomplete reporting, especially in passive surveillance systems, is very common.

Sentinel surveillance uses a pre-arranged **sample** of reporting sources to report all cases of one or more conditions. Usually the sample sources are selected to be those most likely to see cases. Particularly in developing countries, sentinel surveillance provides a practical alternative to population-based surveillance. Under this strategy, health officials define homogenous population subgroups and the regions to be sampled. They then identify institutions that serve the population subgroups of interest, and that can and will obtain data regarding the condition of interest.

9.2. Surveillance Data Source and Analysis

Surveillance systems based on secondary data analysis can make productive use of data sets collected for other purposes. Data collected for marketing surveys, patient management records, police records, and other information sources can be exploited as sources of surveillance data. Such

data may be of lesser quality and timeliness than data collected through systems designed specifically for surveillance.

As with all descriptive epidemiologic data, surveillance data is first analyzed in terms of time, place, and person. Data are analyzed as rates, rather than simply the numbers of cases reported. When delays occur between diagnosis and reporting, we analyze data by the date of onset, rather than the date of the report. A critical step before calculating rates is the identification of the appropriate denominator. Simple tabular and graphic techniques are used initially to display the data, although sophisticated techniques such as cluster and time series analysis and computer mapping may also be used.

Surveillance data may be assessed for changes over time by comparing the number of cases for the current period with the number reported for the same period in each of the last three years. Secular trends, or long-term trends, are usually analyzed by graphing the occurrence of disease by year. Any key events, such as initiation or cessation of a control program, should be noted on the graph. Changes in the surveillance system (such changes in diagnostic criteria, reporting requirements, screening programs, or publicity about the condition) which may influence the appearance of long-term trends should also be indicated on the graph.

The surveillance data should also be analyzed by place. Even when the secular trend reveals no increases in overall incidence, analysis by place may reveal a geographic cluster of cases which deserves investigation. Analysing surveillance data by the characteristics by person variables (age, sex, behavioral risk factors) may also reveals patterns or clues.

There is no single "threshold" above which disease patterns are different enough from the expected to warrant further investigation. The excess necessary to trigger action may depend on the priority assigned to the disease and the interests, capabilities and resources of the ministry or agency. Public, political, or media attention and pressure, however, can sometimes make it necessary to investigate minor variations in disease occurrence which might no otherwise be pursued.

Apparent increases should be treated as real until proven otherwise. However other causes of apparent increases should also be considered, including an increase in the denominator population, improved detection, "batch" reporting, or other changes in the system itself. Surveillance data should be disseminated to those who provide reports, and those who need to know for administrative, program-planning, and decision-making purposes.

Newsletters and other reports of surveillance data can also help to maintain the quality of a surveillance system by providing motivation for continued reporting by health care providers. Like other epidemiologic data, surveillance data should be "information for action", collected only if it is functionally linked with community health programs.

9.3. Evaluation of Surveillance

In the design or evaluation of a surveillance system, proposals or recommendations for modifications should be made with an understanding of the trade-offs (e.g., the impact that efforts to improve representativeness may have on cost). In justifying, designing or evaluating a surveillance system, the following aspects of the system should be assessed:

- 1) The importance to the public health of the health event under surveillance
 - a) incidence and prevalence
 - b) severity (case-fatality or death-to-case ratio)
 - c) mortality (overall and age-specific mortality rates, years of potential life lost)
 - d) health care costs
 - e) potential for spread
 - f) preventability
- 2) The objectives and operation of the system
 - a) the case definition of the health event
 - b) the population under surveillance
 - c) the time period for data collection (weekly, monthly, annually)
 - d) what information is collected (Is it what programs need?)
 - e) the reporting sources
 - f) how data are handled (transfers, delays, confidentiality)
 - g) how data are analyzed (by whom? frequency, thoroughness)
 - h) how data are disseminated
- 3) The system's usefulness
 - a) Action taken to date as a result of the information
 - b) future or potential uses
- 4) Attributes or qualities of the surveillance system
 - a) simplicity
 - b) flexibility (with changes in case definition or funding, to add new diseases)
 - c) acceptability (often judged by proportion who report, completeness of forms)
 - d) sensitivity (ability to detect events it is intended to detect)
 - e) predictive value positive (proportion of reported cases which truly are cases, or of epidemics which are actual epidemics)
 - f) representativeness (extent to which one can generalize or draw conclusions from surveillance data, such as for calculating rates)
 - g) timeliness
- 5) Cost or resource requirements for system operation

Surveillance systems are never perfect. Understanding the limitations of surveillance data is important to ensure correct interpretation. The most common limitations of surveillance systems include:

-
- 1) Under reporting (such as due to lack of knowledge of reporting requirements, negative attitudes toward reporting)
 - 2) Lack of representativeness of reported cases (such as due to a bias toward reporting severe cases, or increased likelihood of reporting after publicity)
 - 3) Lack of timeliness
 - 4) Inconsistency of case-definitions

These limitations suggest specific steps which may be taken to improve a surveillance system. Most commonly, surveillance systems are strengthened by improving awareness of practitioners, simplification of the process of reporting, frequent feedback to those reporting, widening the "net" (for example, obtaining reports from laboratories or schools, rather than relying on physicians), and using active (rather than passive) surveillance. Remember to "share the data, share the responsibility, and share the credit".

Important Points

Factors related with the selection of disease for surveillance:

- Magnitude of the disease
- Feasibility of control measures
- Need for monitoring and evaluating the performance of a control program
- Resource availability

Activities in surveillance:

- Data collection and recording
- Reporting and notification
- Compilation, data analysis, and interpretation
- Dissemination of findings for action

Conditions in which active surveillance is appropriate:

- For periodic evaluation of ongoing programs
E.g. HIV/AIDS, EPI...
- For programs which have time limit of operation
E.g. Small pox
- With the occurrence of unusual situations:
 - ☐ when a new disease/event discovered
 - ☐ when investigating a new mode of transmission
 - ☐ when a high-risk period is recognized
 - ☐ when a disease appears in a new geographic area or found to affect a new subgroup of the population
 - ☐ when previously eradicated disease reappear or low incidence disease occur at a higher level of endemicity

Features of good surveillance system

- Uses a combination of passive and active mechanisms to collect data.
 - . Emphasize the collection of minimum data in a simplest possible way.
 - . To assure quality and enhance compliance make sure that the data collected is useful for the workers who collect the data.
- Timely reporting.
- Timely and comprehensive action.

Action must be targeted towards both case detection and treatment and as well as to the control of the disease.

- Strong laboratory services for accurate diagnosis.

9.4 Integrated Disease Surveillance and Response (IDSR): Concept and Experience in Ethiopia

Integrated disease surveillance and response (IDSR) is an approach adapted to strengthen national disease surveillance systems by coordinating and streamlining all surveillance activities and ensuring timely provision of surveillance data to all disease prevention and control programmes in order to initiate timely response (intervention).

IDSR initiative was launched by the WHO-AFRO (Africa regional office for WHO) in the second half of the 1990's. Since then the initiative has been adapted by many African countries including Ethiopia. In fact, Ethiopia was one of the countries in Africa that has made good progress in IDSR implementation. Adaptation of the national guidelines and training modules for IDSR, training for professionals from national to woreda level, and preparation and distribution of relevant forms are completed. Data collection and reporting using the IDSR guideline and forms is also initiated.

Experiences from disease eradication and elimination programs in developing countries show that disease control and prevention objectives are successfully met when resources are dedicated to improving the ability of health systems to detect targeted diseases, obtain laboratory confirmation of epidemics, and use action thresholds at the woreda level in an integrated fashion. Improving communicable disease surveillance and response through integrated disease surveillance (IDSR) linking community, health facility, woreda and national levels in the country promotes rational use of resources. The use un-integrated disease surveillance systems involving using the same constrained structures, processes and personnel puts a lot of unnecessary pressure on the system and can not produce results as needed and effectively. Thus, integration of the surveillance and response activities offers a lot of advantages.

Integrated disease surveillance and response:

- Focuses at the woreda level, as this is the lowest level in the health system with full-time staff dedicated to all aspects of public health.
- Coordinates and streamline all surveillance activities combining available resources (human, material, financial...) from a single focal point at woreda level.

- Facilitate collaboration between surveillance focal points at the woreda, regional and national levels and epidemic response committees at each level in taking appropriate and timely public health responses and actively seek opportunities for combining resources.

The overall objective of the IDSR is to improve the ability of health workers to detect and respond to priority communicable diseases at the woreda level. Effective and timely decision-making based on good evidence increases efficient utilization of available resources for preventing and controlling communicable diseases and improving the health status of the population.

IDSR in order to achieve its objectives seeks to:

- Strengthen the capacity of woredas to conduct effective surveillance activities
- Integrate multiple surveillance systems so that forms, personnel and resources can be used more efficiently and effectively
- Improve the use of information for decision making
- Improve the flow of surveillance information between and within levels of the health system
- Improve laboratory capacity in identification of pathogens and monitoring of drug sensitivity
- Increase the involvement of health workers in the surveillance system.
- Emphasize community participation in detection and response to public health problems
- Strengthen the involvement of laboratory personnel in epidemiological surveillance.

One of the first activities in the implementation of IDSR was to orient health professional on the initiative and provide the necessary theoretical as well as practical skills needed for implementation through a series of training. The training module is developed based on the experience and data derived from actual management of epidemic diseases across Africa. The training module provides detailed guidance on how to maintain a good surveillance system that is relevant to offer effective response- detection of target diseases, analysis and interpretation of surveillance data, investigation of and response to suspected epidemics, epidemic preparedness, and supervision of surveillance activities. Students are advised to refer to the national guidelines to gain insight on the priority diseases for surveillance.

X. SCREENING

Screening is a public health intervention intended to improve the health of a precisely defined target population.

Screening is the presumptive identification of unrecognized disease or defect by the application of tests, examinations or other procedures which can be applied rapidly. Screening tests sort out apparently well persons who probably have a disease from those who probably do not. A screening test is not intended to be diagnostic. (Last 2001)

The purpose of health interventions is to reduce severe morbidity and mortality by shifting the natural history of the disease to the right and making disease less severe. One way of achievement detection of diseases at early stage of its natural course is by applying a screening test.

The aim of screening program includes:

- To reverse, halt, or slow the progression of disease more effectively than would probably normally happen.
- To alter the natural course of disease for a better outcome for individuals affected.
- Protect society from contagious disease
- Rational allocation of resources
- Selection of healthy individuals: employment, military...
- Research; study on natural history of diseases

In the belief that 'prevention is better than cure' there has been widespread enthusiasm for screening populations for illness in its early stage, so that a better outcome can be achieved by more effective intervention. This concept has grown in strength in western populations because of the increasing importance of cancer as a major cause of death. Cancer with its insidious course and natural history which moves from a localised and treatable phase to a widespread and untreatable one is an example of a group of conditions for which screening has been believed to be appropriate. Identification of those who may have a problem and who might benefit from further investigation and treatment is the core of a screening program. Thus, It involves the application of a quick and simple test, usually by paramedics, to large numbers of normal persons, so that those with a possible problem can be identified as early as possible.

For example, accurate early diagnosis of cancer (or pre-cancer) gives the opportunity to start treatment before disease progresses, thus potentially

reducing the need for aggressive therapy, reducing the likelihood of metastatic disease, and averting cancer deaths.

It is important to note that screening is potentially expensive and it is an intervention which is thrust upon the public rather than a response to an individual seeking help. It must, therefore, be constantly and carefully monitored both for its processes and its effectiveness. False negative tests are a constant source of concern and there is often public outrage after such occurrences. False positive tests cause undue anxiety and wasted resources. But for the screening programme to be effective, it must reach a high proportion of the population at risk. Failure to achieve this leads to the failure of the programme to meet its targets. This is measured by coverage - the proportion of the target population successfully tested in each screening activity. Keep in mind that it is those often hardest to reach in screening programmes who suffer from the worst disease.

Screening test results are often reported as simply normal or abnormal, although some may be more or less abnormal (or normal) than others. The following table summarizes the four possible relationships between a diagnostic test and the actual presence of disease (Table 10.1)

Table 10.1. Results of Screening Testing

Test Result	Result for a Gold Standard		Total
	Disease Present	Disease Absent	
Positive	A	b	$a + b$
Negative	C	d	$c + d$
Total	$a + c$	$b + d$	$a + c + b + d$

Values are defined as follows: a = true-positive results, b = false-positive results, c = false-negative results, and d = true-negative results. Sensitivity is defined as $a/(a + c)$, while specificity is defined as $d/(b + d)$. The positive predictive value is defined as $a/(a + b)$, and the negative predictive value is defined as $d/(c + d)$.

The assessment of what is "true" or "false" depend on the selection of a "gold standard". Although the truth or falsehood of measures by the "gold standard" method may, themselves, be questioned, these are usually the best available information, which is the basis for evaluation of the performance of a second diagnostic test which is usually cheaper, easier, or safer. The ability of the screening test to differentiate between those who are disease free from those who are affected is called the test validity. The screening test validity is measured by sensitivity and specificity.

Sensitivity is defined as the proportion of people with a disease who have a positive test for the disease ($a/a+c$).

Specificity is the proportion of people without the disease who have a negative test ($d/b+d$).

A highly sensitive test is preferable when there is an important penalty for failing to detect a disease (e.g., when trying to detect a dangerous but treatable condition). Sensitive tests are also used when the probability of disease is relatively low and the purpose of the test is to discover possible cases. A sensitive test is, therefore, most helpful when the test result is negative. Specific tests, on the other hand, are most useful when the test result is positive, and are often used to confirm a diagnosis which has been suggested by other data. A highly specific test is preferable when false positive results might have negative (physical, emotional, or financial) consequences.

Case definitions used in epidemiology may also be characterized by their sensitivity and specificity. For rare but potentially severe communicable diseases, where it is important to identify every possible case, health officials use a **sensitive** or "loose" case definition. On the other hand, investigators of the causes of a disease outbreak want to be certain that any person included in the investigation really had the disease. In this case, the investigator prefers a **specific** or "strict" case definition.

In theory, the sensitivity and specificity of a test are independent of the prevalence of the condition being detected. In practice, however, several characteristics of cases (such as the stage and severity of the disease) may be related to both the sensitivity and specificity of a test and to the prevalence of the disease, since different kinds of cases are found in high- and low-prevalence situations. Tests are often assessed to be more valuable than they actually are, since a positive test result may prompt the health care provider to continue pursuing a diagnosis, while a negative result may cause a clinician to abandon further testing.

When assessing the implications of a positive or negative test, the sensitivity and specificity (which are more useful in deciding whether to perform the test) are no longer of primary importance.

Positive predictive value (or predictive value positive) (+PV = $a/a+b$) is the probability of disease in a person with a positive (abnormal) test result.

Negative predictive value (or predictive value negative) (-PV = $d/c+d$) is the probability of not having the disease when the test result is negative (normal). Predictive value is sometimes called *posterior* or *post-test* probability.

The predictive value of a test is not a property of the test alone. It is determined by the sensitivity and specificity of the test and the prevalence of the disease in the population being tested (Figure 10.2). Positive results, even for a very specific test, when applied to a population with a low likelihood of disease, will be largely false positives. Similarly, negative results, even for a very sensitive test, will be largely false negatives when the test is performed in a population with a high chance of having the disease.

Table 10.2. Effect of Prevalence on the Positive Predictive Value, with 90% Sensitivity and 95% Specificity

Prevalence, %	Positive Predictive Value, %
0.1	1.80
1.0	15.4
5.0	48.6
50.0	94.7

The criteria for a successful screening program were first summarized in a WHO publication in 1968. They can be broadened to screening for problems other than human disease:

1. The problem to be detected should be important enough to be worth detecting.
2. There should be an acceptable intervention which is effective.
3. The intervention should be feasible and available.
4. There should be a recognizable latent or early "asymptomatic" stage.
5. There should be a suitable test.
6. The test should be acceptable to the population to be tested.
7. The natural history of the condition should be adequately understood.
8. There should be an agreed policy regarding when the intervention is appropriate.
9. The cost of detecting the problem and its remedy should be reasonable.
10. The screening program should be ongoing, and not a "one-time" effort.

Screening Test

Laboratory tests for screening are used in people who are asymptomatic (apparently healthy individuals) to classify their likelihood of having a particular disease. A test is anything that produces evidence from a patient at any stage in the clinical process, based on which a different clinical course will be taken depending on the different possible test outcomes (positive or negative, normal or abnormal, present or absent, high or low, ...). The screening procedure is not the only basis for the diagnosis of illness. Patients with positive test results are referred for subsequent testing or

examination to provide the physician with more information to determine if they have the disease in question.

A disease should be serious to warrant large-scale screening for it, and treatment before symptoms develop or deteriorate should be of more benefit in reducing morbidity and mortality than treatment later. The estimated prevalence of preclinical disease should be high in the population being screened. Once these criteria have been met, the issue is examined from the standpoint of laboratory tests.

An acceptable screening test is one that is highly accurate, i.e., results are positive for almost all individuals with the disease, and the physician can be confident that the patient is actually free of the disease when test results are negative. Specificity is important when one is screening for rare diseases because false-positive results are possible when the test is not specific. The basic tenets of decision analysis indicate that a particular intervention is undertaken when benefits outweigh costs. Therefore, the ideal screening test is inexpensive, easy to administer, and poses little risk and causes minimal discomfort for the patient. In addition, results of the screening test must be valid, reliable, and reproducible.

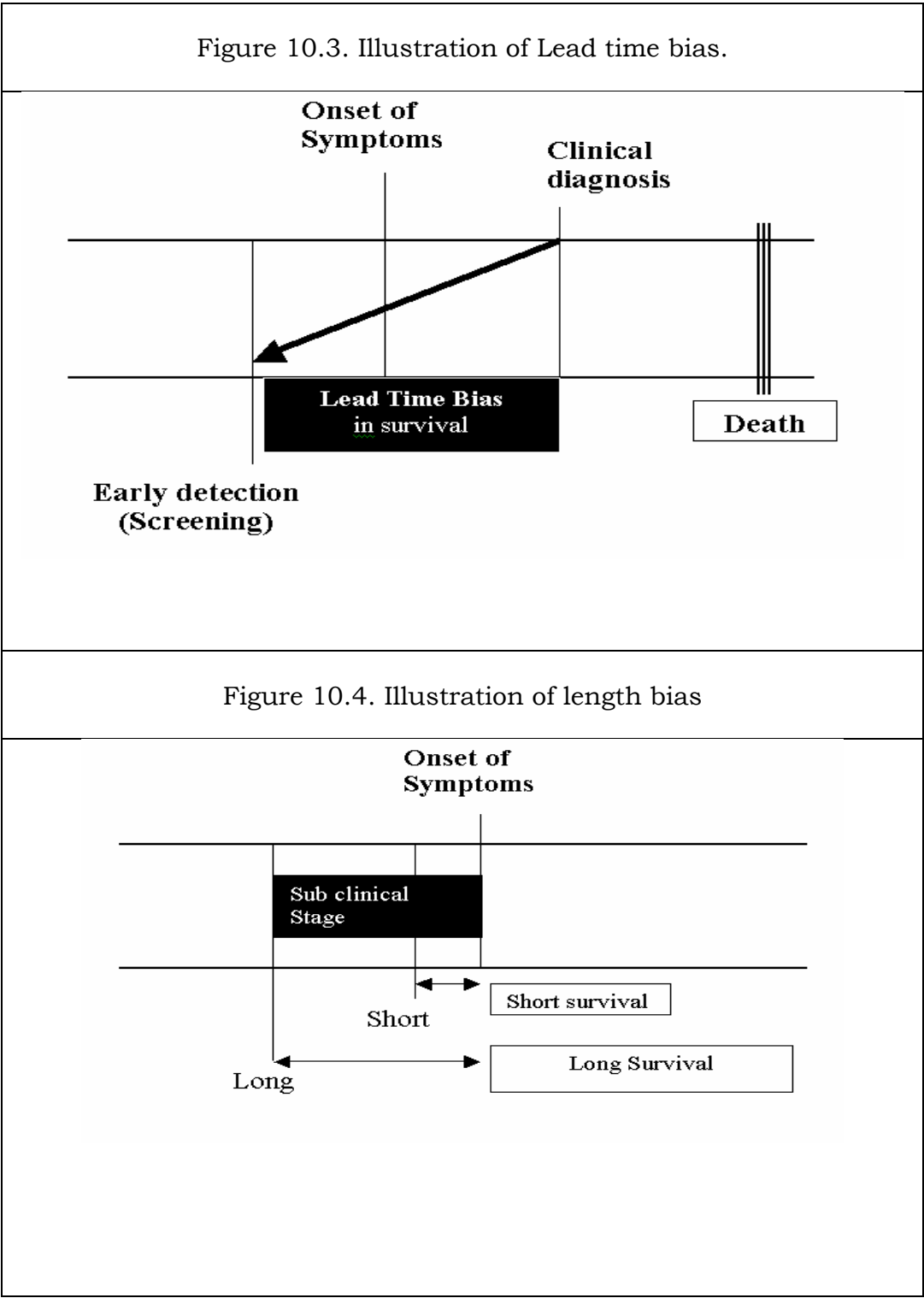
Potential Source of Bias in Screening include

- Self-selection (volunteer) bias: this implies that those accepting screening are different from those declining it. Comparing populations accepting and declining for all relevant characteristics.

- Lead time bias (early diagnosis): this is a bias caused by picking screened cases at an early stage of the disease, i.e., before they develop signs and symptoms of the disease (Figure 10.3). There are two ways for accounting such differences:

1. Adjust survival data for estimated lead time
2. Stage disease and compare morbidity/mortality within stages

- Length Bias (chronicity and progression): this is related to the variation in the speed of progression of the disease (Figure 10.4). Cases picked up by a screening may be less severe, and slow progressive compared with others. It is very important to be aware of such a possibility in interpreting survival gains from a screening program.



XI. ETHICS OF EPIDEMIOLOGIC RESEARCH

Epidemiologic research, with its continually expanding potential for collection, storage and use of data on individuals and communities, encounters inevitable conflicts between the rights and freedoms of the individual and the needs of society. Past ethical abuses, particularly in clinical research, have underlined the need for clear guidelines for the ethical conduct of both clinical and epidemiologic research. The *Proposed International Guidelines for Biomedical Research Involving Human Subjects* were adopted by the World Medical Association in 1964 to guide application of the ethical principles for clinical research specified in the Declaration of Helsinki. The additional need for special ethical guidelines for epidemiologic studies has also recently been accentuated by the complex issues raised by research regarding HIV infections and AIDS.

Relevant International Ethical Guidelines are:

- **1947**, the Nuremberg Code
- **1964**, The Declaration of Helsinki (World Medical Association)...revised in 1975, 1983, 1989, 1996, 2000, 2002
- **1991**, *International Guidelines for Ethical Review of Epidemiological Studies* (CIOMS= Council for International Organizations of Medical Sciences)
- **1993**, *International Ethical Guidelines for Biomedical Research Involving Human Subjects* (CIOMS)... revised in 2002

As can be observed from the above list of guidelines, the frequency of revisions has increased remarkably in the past two decades. Some of the major reasons for the frequent revisions are:

- HIV/AIDS pandemic; initiation of large-scale trials of vaccine and treatment drugs to prevent the spread of disease and provide treatment for people living with HIV/AIDS.
- Rapid advances in medicine and biotechnology
- Changing research practices such as multinational field trials
- Experimentation involving vulnerable population groups such as pregnant women, children and mentally ill individuals.
- Involvement of populations in developing countries in human experimentation.

The Council for International Organizations of Medical Sciences (CIOMS) in collaboration with the World Health Organization (WHO) published the first "International Guidelines for the Ethical Review of Epidemiological Studies". The following discussion of epidemiologic research ethics draws largely on the guidelines. Although such ethical guidelines cannot resolve all the moral ambiguities that are

encountered in everyday epidemiologic research and practice, they can draw attention to the ethical implications of professional action and thereby improve ethical standards. It is also important to be aware of the national ethical guideline that in fact governs the research practice in the country. The national guidelines in addition to endorsing the international guidelines provide some clear directives on issues that are more pertinent locally.

Ethical issues often arise as a result of conflict among competing sets of values. Many situations require careful discussion and informed judgements on the part of investigators, ethical review committees, administrators, health care practitioners, policy-makers, and community representatives. Externally sponsored epidemiological studies in developing countries merit special attention in ethical review.

The purpose of ethical review is to consider the features of a proposed study in light of ethical principles, so as to ensure that investigators have anticipated and satisfactorily resolved possible ethical objections, and to assess their response to ethical issues raised by the study. Not all ethical principles weigh equally. A study may be assessed as ethical even if a usual ethical expectation, such as confidentiality of data, has not been comprehensively met, provided the potential benefits clearly outweigh the risks and the investigators give assurances of minimizing risks. It may even be unethical to reject such a study, if its rejection would deny a community the benefits it offers. The challenge of ethical review is to take into account potential risks and benefits, and to reach decisions which best reflect the consensus of the review committee. Different conclusions may result from different ethical reviews of the same issue or proposal, and each conclusion may be ethically reached, given varying circumstances of place and time; a conclusion is ethical not merely because of what has been decided, but also owing to the process of conscientious reflection and assessment by which it has been reached.

11.1 General Ethical Principles

General ethical principles may be applied at the individual and community levels. At the level of the individual (microethics), ethics governs how one person should relate to another and the moral claims of each member of a community. At the level of the community, ethics applies to how one community relates to another, and to how a community treats each of its members (including prospective members) and members of other groups with different cultural values (macroethics). Procedures that are unethical at one level cannot be justified merely because they are considered ethically acceptable at the other.

All research involving human subjects should be conducted in

accordance with four basic ethical principles: 1) respect for persons, 2) beneficence, 3) non-maleficence, and 4) justice.

Respect for persons incorporates at least two other fundamental ethical principles, namely:

- a) autonomy, which requires that those who are capable of deliberation about their personal goals should be treated with respect for their capacity for self-determination; and,
- b) protection of persons with impaired or diminished autonomy, which requires that those who are dependent or vulnerable be afforded security against harm or abuse.

Beneficence is the ethical obligation to maximize possible benefits and to minimize possible harms and wrongs. This principle gives rise to norms requiring that the risks of research be reasonable in the light of expected benefits, that the research design be sound, and that the investigators be competent both to conduct the research and to assure the well-being of the research subjects.

Non-maleficence ("Do no harm") holds a central position in the tradition of medical ethics, and guards against avoidable harm to research subjects.

Justice requires that cases considered to be alike be treated alike, and that cases considered to be different be treated in ways that acknowledge the difference. When the principle of justice is applied to dependent or vulnerable subjects, its main concern is with the rules of distributive justice. Studies should be designed to obtain knowledge that benefits the class of persons of which the subjects are representative. The class of persons bearing the burden should receive an appropriate benefit, and the class primarily intended to benefit should bear a fair proportion of the risks and burdens of the study.

11.2. Ethical Principles Applied to Epidemiology

1. Informed Consent

When individuals are the subject of epidemiologic studies, their individual informed consent will usually be sought. Consent is informed when it is given by a person who understands the purpose and nature of the study, what participation in the study requires the person to do and to risk, and what benefits are intended to result from the study. An investigator who proposes not to seek informed consent has the obligation to explain how the study would be ethical in its absence (such as because informed subjects might alter the behaviour under study or feel needlessly anxious or because it is public

knowledge that personal data is made available for epidemiologic studies). Consent is not required for use of publicly available information, although countries and communities differ with regard to the definition of what information about citizens is regarded as public. Investigators must provide assurances that strict safeguards will be maintained to protect confidentiality by minimizing disclosure of personally sensitive information.

When it is not possible to obtain informed consent from every individual to be studied, community agreement through a representative of a community or group may be sought, but the representative should be chosen according to the nature, traditions and political philosophy of the community or group. Approval given by a community representative should be consistent with general ethical principles. Even if a leader expresses agreement on behalf of a community, the refusal of individuals to participate has to be respected. Representatives of a community or group may sometimes be invited to participate in the design of a study and in its ethical assessment.

Selective disclosure may be used in epidemiologic research, provided that it does not induce subjects to do what they would not otherwise consent to do. For certain epidemiologic studies, such non-disclosure is permissible, even essential, so as not to influence the spontaneous conduct under investigation, and to avoid obtaining responses that the respondent might give in order to please the questioner.

Prospective subjects may not feel free to refuse requests from those who have power or undue influence over them. It is ethically questionable whether subjects should be recruited from among groups that are unduly influenced by persons in authority if the study can be conducted with subjects who are not in this category.

Individuals or communities should not be pressured to participate in a study. However, it can be hard to draw the line between exerting pressure or offering inappropriate inducements to participate and creating legitimate motivation. The benefits of a study, such as improved knowledge or health, are appropriate inducements. However, when people or communities lack basic health services or money, the prospect of being rewarded by goods, services or cash payments can induce participation. It is acceptable to repay incurred expenses, such as for travel.

2. Maximizing Benefit

Part of the benefit that communities, groups, and individuals may reasonably expect from participating in studies is that they will be told of findings that pertain to their health. A strategy for communication of study results to policy-makers and to participating individuals and communities (with due consideration of levels of literacy and comprehension) should be included in the study protocol. When findings indicate a need for health care, those concerned should be appropriately advised and arrangements should be made for treatment or referral. Health professionals have an obligation to advocate release of study results that is in the public interest. Training of local health personnel in skills and techniques that can be used to improve health services or research may also be an important way of ensuring that communities will benefit from the proposed research.

3. Minimizing Harm

Epidemiologic studies must consider all harm or disadvantage to individuals and communities which may be incurred due to the research. For example, diversion of scarce health personnel from their routine duties to serve the needs of a study or alteration of health care priorities may constitute harm. Ethical review should also assess the risk of subjects or groups suffering stigmatization, prejudice, loss of prestige or self-esteem (e.g., due to being identified as HIV-positive), or economic loss as a result of taking part in a study. Investigators must be able to demonstrate that benefits outweigh the risks for both individuals and groups. When a healthy person is a member of a population or sub-group at increased risk and engages in high-risk activities, it is unethical not to propose measures for protecting the population or sub-group.

Disruption of social mores is usually regarded as harmful. Although investigators must respect social mores, it may be the specific aim of an epidemiologic study to stimulate change in certain customs or behaviours to improve health. Investigators must respect the ethical standards of their own country or culture as well as the cultural expectations of the societies in which epidemiological investigations are undertaken.

4. Confidentiality

Research may involve collecting and storing data relating to individuals and groups, and such data, if disclosed to third parties, may cause harm or distress. Consequently, investigators should make arrangements for protecting the confidentiality of such data by, for example, omitting information that might lead to identification of

individual subjects, or limiting access to the data, or by other means. When personal identifiers remain on records used for a study, investigators should explain why this is necessary and how confidentiality will be protected.

Unlinked information is that which cannot be linked, associated or connected with the person to whom it refers. As this person is not known to the investigator, confidentiality is not at stake and the question of consent does not arise. Linked information may be 1) anonymous (when the information cannot be linked to the person to whom it refers except by a code or other means known only to that person, and the investigator cannot know the identity of the person), 2) non-nominal (when the information can be linked to the person by a code which is not a personal identifier and which is known to the person and the investigator), and 3) nominal or nominative (when the information is linked to the person by means of personal identification, usually the name).

5. Conflict of Interest

It is an ethical rule that investigators should have no undisclosed conflict of interest with their study collaborators, sponsors, or subjects. Conflict can arise when a commercial or other sponsor may wish to use study results to promote a product or service, or when it may not be politically convenient to disclose findings. Honesty and impartiality are essential in designing and conducting studies, and presenting and interpreting findings. Data must not be withheld, misrepresented or manipulated.